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CHAPTER INDEX

CHAPTER	SYNOPSIS QR CODE	VSAQ P.No	SAQ P.No	LAQ P.No	OMQ P.No
1. RELATIONS & FUNCTIONS		96 - 97	54 - 55		131
2. INVERSE TRIG.FUNCTIONS			56 - 59		132
3 MATRICES		98-102	60 - 66		133
4. DETERMINANTS		103-104	67 - 72	16-18	134
5. CONTINUITY & DIFFERENTIABILITY		115-117	80 -82		135
6. APPLICATION OF DERIVATIVES		118-119		30 - 34	136
7. INTEGRATION		120-124	83 - 89	35 - 43	137
8. APPLICATION OF INTEGRALS			90 - 92		138
9. DIFFERENTIAL EQUATIONS		125-127		44 - 51	139
10. VECTOR ALGEBRA		105-108	73 - 74	19 - 21	140
11. THREE DIMENSIONAL GEOMETRY		109-111		22 - 25	
12. LINEAR PROGRAMMING			78 - 79		141
13. PROBABILITY		112-114	75 - 77	26 - 29	142

Mahesh

☞ To get **'Synopsis & Math Beats'** of respective chapters please scan the above **'QR Code'** through **Google Lens**

NEW PATTERN (2026-27)
BLUE PRINT of MATHS Q' Paper

CHAPTER (Total Marks)	OMQ(1M) 12x1=12	VSAQ(2M) 10x2=20	SAQ(4M) 7x4=28	LAQ(8M) 5x8=40	Total 100 M
1. Relations and Functions (7)	1	2	4	0	7
2. Inverse Trig. Functions (5)	1	0	4	0	5
3. Matrices (7)	1	2	4	0	7
4*. Determinants (15)	1	2	4	8	15
5. Continuity & Differentiability (7)	1	2	4	0	7
6*. Applications of Derivatives (11)	1	2	0	8	11
7*. Integrals (23)	1	2	4	8+8	23
8. Applications of Integrals (5)	1	0	4	0	5
9*. Differential Equations (11)	1	2	0	8	11
10*. Vector Algebra (15)	1	2	4	8	15
11*. 3D Geometry (10)	0	2	0	8	10
12. Linear Programming (5)	1	0	4	0	5
13*. Probability (15)	1	2	4	8	15
Section wise Marks/ No.of Q's	12M/12Q	20 M/10Q	28 M/7Q	40 M/5Q	100/32Q

Mahesh

* *This Blue print is prepared as per the 'Board Model Papers' issued by B.I.E.*

* indicates LAQ Chapters.

SECTION INDEX

I. LAQ CHAPTERS (15-52)

1.	DETERMINANTS	(8 Marks)	16 - 18
2.	VECTOR ALGEBRA	(8 Marks)	19- 21
3.	3D GEOMETRY	(8 Marks)	22 - 25
4.	PROBABILITY	(8 Marks)	26 - 29
5.	APPLICATION OF DERIVATIVES	(8 Marks)	30 - 34
6.	INTEGRALS	(8 + 8 Marks)	35 - 43
7.	DIFFERENTIAL EQUATIONS	(8 Marks)	44- 51
•	STAR QUESTIONS PLUS	(SQP)	52

II. SAQ CHAPTERS (53-94)

1.	RELATIONS & FUNCTIONS	(4 Marks)	54 - 55
2.	INVERSE & TRIG. FUNCTIONS	(4 Marks)	56 - 59
3.	MATRICES	(4 Marks)	60 - 66
4.	DETERMINANTS	(4 Marks)	67 - 72
5.	VECTOR ALGEBRA	(4 Marks)	73 - 74
6.	PROBABILITY	(4 Marks)	75 - 77
7.	LINEAR PROGRAMMING	(4 Marks)	78 - 79
8.	CONTINUITY &DIFFERENTIABILITY	(4 Marks)	80 - 82
9.	INTEGRALS	(4 Marks)	83 - 89
10.	APPLICATION OF INTEGRALS	(4 Marks)	90 - 92
•	STAR QUESTIONS PLUS	(SQP)	93-94

III.	VSAQ CHAPTERS	95 - 129
IV.	OMQ CHAPTERS	130 - 146
V.	4 MODEL PAPERS(1 BMP + 4 GUESS)	147- 158
•	40 DAYS ACTION PLAN	159- 160

☞ The Chapters are Renumbered as per VSAQ, SAQ, LAQ division.

Maresh

LAQ HIT LIST
TOP 30 LAQ

★★★ DETERMINANTS [8 MARKS] ★★★

LAQ 1. Solve the following system of equations by matrix method.
$3x - 2y + 3z = 8,\ 2x + y - z = 1,\ 4x - 3y + 2z = 4$

LAQ 2. Solve the system of equations using matrix method $2x + 3y + 3z = 5,$
$x - 2y + z = -4,\ 3x - y - 2z = 3$

★★★ VECTOR ALGEBRA [8 MARKS] ★★★

LAQ 3. If $\vec{a}=2\hat{i}+\hat{j}-3\hat{k},\ \vec{b}=\hat{i}-2\hat{j}+\hat{k}, \vec{c}=-\hat{i}+\hat{j}-4\hat{k}$ and $\vec{d}=\hat{i}+\hat{j}+\hat{k}$
then compute $|(\vec{a}\times\vec{b})\times(\vec{c}+\vec{d})|$

LAQ 4. If $\vec{a}=\hat{i}-2\hat{j}-3\hat{k},\ \vec{b}=2\hat{i}+\hat{j}-\hat{k}, \vec{c}=\hat{i}+3\hat{j}-2\hat{k}$ then, verify $\vec{a}\times(\vec{b}\times\vec{c}) \neq (\vec{a}\times\vec{b})\times\vec{c}$

LAQ 5. If $\vec{a}=\hat{i}-2\hat{j}+3\hat{k},\ \vec{b}=2\hat{i}+\hat{j}+\hat{k},\ \vec{c}=\hat{i}+\hat{j}+2\hat{k}$ then find $|(\vec{a}\times\vec{b})\times\vec{c}|$ and $|\vec{a}\times(\vec{b}\times\vec{c})|$

★★★ 3D [8 MARKS] ★★★

LAQ 6. Find the shortest distance between the lines
$\vec{r}=(\hat{i}+2\hat{j}+\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=2\hat{i}-\hat{j}-\hat{k}+\mu(2\hat{i}+\hat{j}+2\hat{k})$

LAQ 7. Find the shortest distance between the lines

$$\frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1} \quad \text{and} \quad \frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1}$$

LAQ 8. Find the distance between the lines l_1 and l_2 given by
$\vec{r}=\hat{i}+2\hat{j}-4\hat{k}+\lambda(2\hat{i}+3\hat{j}+6\hat{k}),\ \vec{r}=3\hat{i}+3\hat{j}-5\hat{k}+\mu(2\hat{i}+3\hat{j}+6\hat{k})$

★★★ PROBABILITY [8 MARKS] ★★★

LAQ 9. Bag I contains 3 red and 4 black balls while another Bag II contains 5 red and 6 black balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from Bag II.

LAQ 10. Given three identical boxes I, II and III, each containing two coins. In box I, both coins are gold coins, in box II, both are silver coins and in the box III, there is one gold and one silver coin. A person chooses a box at random and takes out a coin. If the coin is of gold, what is the probability that the other coin in the box is also of gold?

LAQ 11. In a factory which manufactures bolts, machines A, B and C manufacture respectively 25%, 35% and 40% of the bolts. Of their outputs, 5, 4 and 2 percent are respectively defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it is manufactured by the machine B?

Ans-Pag

[☞P 16(1)]

[☞P 17(2)]

[☞P 19(6)]

[☞P 20(7)

[☞P 21(9)

[☞P 22(1

[☞P 24(

[☞P 24(P

[☞P 26(1

[☞P 27(2

[☞P 27(

Mahesh

★★ APPLICATION OF DERVIATIVES [8 MARKS] ★★

LAQ 12. Sand is pouring from a pipe at the rate of 12 cm^3/s. The falling sand forms a cone on the ground in such a way that the height of the cone is always one-sixth of the radius of the base. How fast is the height of the sand cone increasing when the height is 4 cm? |☞P 30(25)|

LAQ 13. Find two positive numbers x and y such that x+y=60 and xy^3 is maximum. |☞P 31(28)|

LAQ 14. Show that of all the rectangles inscribed in a given fixed circle, the square has the maximum area. |☞P 32(29)|

LAQ 15. A rectangular sheet of tin 45 cm by 24 cm is to be made into a box without top, by cutting off square from each corner and folding up the flaps. What should be the side of the square to be cut off so that the volume of the box is maximum ? |☞P 33(31)|

★★★ INTEGRALS [8 + 8 MARKS] ★★★

LAQ 16. Evaluate $\int_0^{\pi} \frac{x\,dx}{1+\sin x}$ |☞P 35(34)|

LAQ 17. Evaluate $\int_0^{\pi/4} \log(1+\tan x)\,dx$ |☞P 36(36)|

LAQ 18. Evaluate $\int_0^{1} \frac{\log(1+x)}{1+x^2}\,dx$ |☞P 36(37)|

LAQ 19. Evaluate $\int_0^{\pi/4} \frac{\sin x + \cos x}{9 + 16\sin 2x}\,dx$ |☞P 37(38)|

LAQ 20. Find integral of $\frac{x^2+x+1}{(x+1)^2(x+2)}$ |☞P 38(40)|

LAQ 21. Find $\int \frac{x^2+x+1\,dx}{(x+2)(x^2+1)}$ |☞P 38(41)|

LAQ 22. Find $\int \frac{x^2}{(x^2+1)(x^2+4)}\,dx$ |☞P 39(43)|

LAQ 23. Find integral of $\frac{x+3}{x^2-2x-5}$ |☞P 41(45)|

★★ DIFFERENTIAL EQUATIONS [8 MARKS] ★★

LAQ 24. Solve the differential equation $(\tan^{-1}y - x)\,dy = (1+y^2)\,dx$. |☞P 44(49)|

LAQ 25. Find the general solution of $(1+x^2)dy + 2xydx = \cot x\,dx,\ (x \neq 0)$ |☞P 44(50)|

LAQ 26. Find the general solution of $x\frac{dy}{dx} + y - x + xy\cot x = 0,\ (x \neq 0)$ |☞P 45(51)|

LAQ 27. Find the particular solution of the differential equation $\frac{dy}{dx} + y\cot x = 2x + x^2\cot x (x \neq 0)$ given that y = 0 when $x = \frac{\pi}{2}$ |☞P 46(53)|

LAQ 28. Find the equation of a curve passing through the origin given that the slope of the tangent to the curve at any point (x, y) is equal to the sum of the coordinates of the point. |☞P 47(55)|

LAQ 29. Show that the differential equation $(x^2 - y^2)\,dx + 2xy\,dy = 0$ is homogeneous and solve it. |☞P 48(56)|

LAQ 30. Find the particular solution of $x^2\,dy + (xy + y^2)\,dx = 0$; y = 1 when x = 1 |☞P 50(59)|

Mahesh

SAQ HIT LIST
TOP 40 SAQ

Mahesh

		Ans-Page
	★★ RELATIONS &FUNCTIONS [4MARKS]★★	
SAQ 1.	Determine whether the Relation R in the set A = {1, 2, 3, 4, 5, 6} as R = {(x, y): y is divisible by x} is reflexive, symmetric and transitive.	[☞P 54(61)]
SAQ 2.	State whether the function f : R → R defined by f (x) = 3 – 4x is one-one, onto or bijective. Justify your answer.	[☞P 55(64)]
	★★★ INVERSE TRIG. FUNCTIONS [4MARKS]★★★	
SAQ 3.	Prove that $\cos^{-1}\frac{12}{13}+\sin^{-1}\frac{3}{5}=\sin^{-1}\frac{56}{65}$	[☞P 56(66)]
SAQ 4.	Prove that $\sin^{-1}\left(\frac{3}{5}\right)+\cos^{-1}\left(\frac{12}{13}\right)=\cos^{-1}\left(\frac{33}{65}\right)$	[☞P 56(67)]
SAQ 5.	Prove that $\sin^{-1}\frac{8}{17}+\sin^{-1}\frac{3}{5}=\tan^{-1}\frac{77}{36}$	[☞P 58(70)]
SAQ 6.	Prove that $\tan^{-1}\frac{63}{16}=\sin^{-1}\frac{5}{13}+\cos^{-1}\frac{3}{5}$	[☞P 58(71)]
SAQ 7.	Write the function $\tan^{-1}\frac{\sqrt{1+x^2}-1}{x}, x\neq 0$ in the simplest form	[☞P 59(74)]
	★★★ MATRICES [4MARKS]★★★	
SAQ 8.	Express the matrix $\begin{bmatrix}6&-2&2\\-2&3&-1\\2&-1&3\end{bmatrix}$ as the sum of a symmetric and a skew symmetric matrix	[☞P 61(77)]
SAQ 9.	If $A=\begin{bmatrix}0&6&7\\-6&0&8\\7&-8&0\end{bmatrix}$, $B=\begin{bmatrix}0&1&1\\1&0&2\\1&2&0\end{bmatrix}$, $C=\begin{bmatrix}2\\-2\\3\end{bmatrix}$ Calculate AC, BC and (A + B)C. Also, verify that (A + B)C = AC + BC	[☞P 63(79)]
SAQ 10.	If $A=\begin{bmatrix}-1&2&3\\5&7&9\\-2&1&1\end{bmatrix}$ and $B=\begin{bmatrix}-4&1&-5\\1&2&0\\1&3&1\end{bmatrix}$, then verify that $(A+B)'=A'+B'$	[☞P 64(81)]
SAQ 11.	If $A=\begin{bmatrix}1&2&3\\3&-2&1\\4&2&1\end{bmatrix}$, then show that $A^3-23A-40I=O$	[☞P 65(84)]

★★★ DETERMINANTS [4MARKS] ★★★

SAQ 12. Find equation of line joining (1, 2) and (3, 6) using determinants. [☞P 68(90)]

SAQ 13. Find the inverse of the matrix $\begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 0 \\ -7 & 2 & 1 \end{bmatrix}$ if it exists. [☞P 69(93)]

SAQ 14. If $A=\begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$ Verify A (adj A)=(adj A) A=|A| I [☞P 70(94)]

SAQ 15. If $A=\begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}$ and $B=\begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$, then verify that $(AB)^{-1}=B^{-1}A^{-1}$ [☞P 71(96)]

SAQ 16. Show that $\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix}=4xyz$ [☞P 72(98)]

★★★ VECTOR ALGEBRA [4MARKS] ★★★

SAQ 17. Find a unit vector perpendicular to each of the vector $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$, where $\vec{a}=3\hat{i}+2\hat{j}+2\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-2\hat{k}$ [☞P 73(101)]

SAQ 18. If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=2\hat{i}-\hat{j}+3\hat{k}$ and $\vec{c}=\hat{i}-2\hat{j}+\hat{k}$ find a unit vector parallel to the vector $2\vec{a}-\vec{b}+3\vec{c}$. [☞P 73(102)]

SAQ 19. Show that the points $A(2\hat{i}-\hat{j}+\hat{k})$, $B(\hat{i}-3\hat{j}-5\hat{k})$, $C(3\hat{i}-4\hat{j}-4\hat{k})$ are the vertices of a right angled triangle. [☞P 74(104)]

SAQ 20. S.T $A(-2\hat{i}+3\hat{j}+5\hat{k})$, $B(\hat{i}+2\hat{j}+3\hat{k})$ and $C(7\hat{i}-\hat{k})$ are collinear. [☞P 74(PQ)]

★★★ PROBABILITY [4MARKS] ★★★

SAQ 21. Given two independent events A and B such that P(A) = 0.3, P(B) = 0.6. Find (i) P(A and B) (ii) P(A and not B) (iii) P(A or B) (iv) P(neither A nor B) [☞P 75(106)]

SAQ 22. Let A and B be independent events with P (A) = 0.3 and P(B) = 0.4. Find (i) $P(A \cap B)$ (ii) $P(A \cup B)$ (iii) P (A|B) (iv) P (B|A) [☞P 75(107)]

SAQ 23. If $P(A)=\frac{6}{11}$, $P(B)=\frac{5}{11}$ and $P(A\cup B)=\frac{7}{11}$, find (i) $P(A\cap B)$ (ii) P(A|B) (iii) P(B|A) [☞P 75(108)]

SAQ 24. Probability of solving specific problem independently by A and B are $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If both try to solve the problem independently, find the probability that (i) the problem is solved (ii) exactly one of them solves the problem. [☞P 77(112)]

Mahesh

★★ LINEAR PROGRAMMING [4MARKS]★★

SAQ 25. Maximise $Z = 3x + 4y$ subject to the constraints : $x + y \le 4, x \ge 0, y \ge 0$ [☞P 78(114)]

SAQ 26. Minimise $Z = -3x + 4y$ subject to $x + 2y \le 8, 3x + 2y \le 12, x \ge 0, y \ge 0$. [☞P 79(116)]

★★ CONTINUITY & DIFFERENTIABILITY [4MARKS]★★

SAQ 27. Find the relationship between a and b so that the function f defined by [☞P 80(118)]

$$f(x) = \begin{cases} ax+1, & \text{if } x \le 3 \\ bx+3, & \text{if } x > 3 \end{cases}$$ is continuous at $x = 3$.

SAQ 28. Find $\frac{dy}{dx}$ of $y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right), 0 < x < 1$. [☞P 81(120)]

SAQ 29. Differentiate the function $(\log x)^{\cos x}$ w.r.t. x. [☞P 82(124)]

SAQ 30. Find $\frac{dy}{dx}$ of the function $y^x = x^y$ [☞P 82(125)]

★★★ INTEGRATION [4MARKS]★★★

SAQ 31. Evaluate $\int_0^{\pi/2} \frac{\cos^5 x}{\sin^5 x + \cos^5 x} dx$ [☞P 83(126)]

SAQ 32. Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ [☞P 83(127)]

SAQ 33. Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx$ [☞P 84(129)]

SAQ 34. Find the integral of $\frac{3x-1}{(x-1)(x-2)(x-3)}$ [☞P 85(131)]

SAQ 35. Find $\int \frac{1}{1+\tan x} dx$ [☞P 87(136)]

SAQ 36. Find the integral of $\sin x \sin 2x \sin 3x$ [☞P 88(137)]

SAQ 37. Find the integral of $\frac{\sin x}{\sin(x-a)}$ [☞P 89(139)]

★★ APPLICATION OF INTEGRALS [4MARKS]★★

SAQ 38. Find the area cut off between the line $y = 0$ and the parabola $y = x^2 - 4x + 3$ [☞P 91(147)]

SAQ 39. Find the area bounded by the curve $y = \cos x$ between $x = 0$ and $x = 2\pi$. [☞P 91(148)]

SAQ 40. Find the area enclosed by the line $y=3x$ and curve $y=6x-x^2$. [☞P 91(150)]

Mahesh

VSAQ HIT LIST

TOP 55 VSAQ

Ans-Page

★★★RELATIONS & FUNCTIONS [2 MARKS] ★★★

VSAQ 1. Let A = {1, 2, 3}, B = {4, 5, 6, 7} and let f = {(1, 4), (2, 5), (3, 6)} be a function from A to B. Show that f is one-one. [☞P 96(153)]

VSAQ 2. Check the injectivity and surjectivity of the function $f : N \to N$ given by $f(x) = x^3$ [☞P 96(155)]

3. Check the injectivity and surjectivity of the function $f : Z \to Z$ given by $f(x) = x^2$ [☞P 96(157)]

★★★MATRICES[2 MARKS] ★★★

Q 4. If a matrix has 8 elements, what are the possible orders it can have? [☞P 98(163)]

AQ 5. Find X and Y, if $X+Y=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix}$ and $X-Y=\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}$ [☞P 98(165)]

VSAQ 6. Find x and y, if $2\begin{bmatrix}1 & 3\\ 0 & x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$ [☞P 98(166)]

VSAQ 7. If $x\begin{bmatrix}2\\ 3\end{bmatrix}+y\begin{bmatrix}-1\\ 1\end{bmatrix}=\begin{bmatrix}10\\ 5\end{bmatrix}$ find the values of x and y. [☞P 99(167)]

VSAQ 8. Find the values of x,y and z from $\begin{bmatrix}x+y+z\\ x+z\\ y+z\end{bmatrix}=\begin{bmatrix}9\\ 5\\ 7\end{bmatrix}$ [☞P 99(168)]

VSAQ 9. Find the values of x,y and z from $\begin{bmatrix}x+y & 2\\ 5+z & xy\end{bmatrix}=\begin{bmatrix}6 & 2\\ 5 & 8\end{bmatrix}$ [☞P 99(169)]

VSAQ 10. If $A=\begin{bmatrix}\sin\alpha & \cos\alpha\\ -\cos\alpha & \sin\alpha\end{bmatrix}$ then verify that A'A=I [☞P 101(175)]

VSAQ 11. Show that $A=\begin{bmatrix}1 & -1 & 5\\ -1 & 2 & 1\\ 5 & 1 & 3\end{bmatrix}$ is a symmetric matrix. [☞P 102(177)]

VSAQ 12. Show that $A=\begin{bmatrix}0 & 1 & -1\\ -1 & 0 & 1\\ 1 & -1 & 0\end{bmatrix}$ is a skew symmetric matrix. [☞P 102(178)]

Manesh

★★★DETERMINANTS [2 MARKS] ★★★

VSAQ 13. Find the values of x for which $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$ [☞ P 103(183)]

VSAQ 14. If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A| = 4|A|$ [☞ P 103(185)]

VSAQ 15. Find adjA for $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ [☞ P 104(187)]

VSAQ 16. Find the inverse of the matrix $\begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$ [☞ P 104(188)]

★★★VECTOR ALGEBRA [2 MARKS] ★★★

VSAQ 17. Find the unit vector in the direction of the sum of the vectors, $\vec{a} = 2\hat{i} + 2\hat{j} - 5\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} + 3\hat{k}$. [☞P 105(193)]

VSAQ 18. Find a vector in the direction of vector $5\hat{i} - \hat{j} + 2\hat{k}$. which has magnitude 8 units. [☞ P 106(194)]

VSAQ 19. Find $|\vec{x}|$, if for a unit vector $\vec{a}$, $(\vec{x} - \vec{a}).(\vec{x} + \vec{a}) = 12$ [☞ P 106(196)]

VSAQ 20. Show that the points $A(-2\hat{i} + 3\hat{j} + 5\hat{k})$, $B(\hat{i} + 2\hat{j} + 3\hat{k})$, $C(7\hat{i} - \hat{k})$ are collinear. [☞ P 107(PQ)]

VSAQ 21. If $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$, $\vec{c} = 3\hat{i} + \hat{j}$ are such that $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{c}$, then find the value of λ. [☞P 108(204)]

VSAQ 22. Find the area of a parallelogram whose adjacent sides are $\vec{a} = 3\hat{i} + \hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ [☞P 108(207)]

★★★3D GEOMETRY [2 MARKS] ★★★

VSAQ 23. If a line makes an angles of 60° and 45° with the direction of positive X and Z aixs respectively. Find the angle made by the line with the Y-aixs. [☞P 109(209)]

VSAQ 24. Find the direction cosines of a line which makes equal angles with the coordinate axes. [☞P 109(210)]

VSAQ 25. Find the direction cosines of the joining the points (–2, 6, 7) , (1, 2, –5). [☞P 110(211)]

VSAQ 26. Show that the points (2, 3, 4), (– 1, – 2, 1), (5, 8, 7) are collinear. [☞P 110(212)]

VSAQ 27. Find the angle between the following pairs of lines: [☞P 110(215)]

$\vec{r} = 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k})$ and $\vec{r} = 7\hat{i} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$

★★★PROBABILITY [2 MARKS] ★★★

VSAQ 28. Compute P(A|B), if P(B) = 0.5 and P (A ∩ B) = 0.32 [☞ P 112(219

VSAQ 29. If $P(A) = \frac{3}{5}, P(B) = \frac{1}{5}$, find P (A ∩ B) if A and B are independent events. [☞ P 112(221

VSAQ 30. Given that E and F are events such that P(E) = 0.6, P(F) = 0.3 and P(E ∩ F) = 0.2, find P(E|F) and P (F|E) [☞ P 112(223

VSAQ 31. Determine P(E|F) when a coin is tossed three times, where E : at least two heads , F : at most two heads [☞P 113(225)]

Mahesh

★★★CONTINUITY & DIFFERENTIABILITY [2 MARKS] ★★★

VSAQ 32. Differentiate the function cos(sin x) with respect to x . [☞P 115(232)]

VSAQ 33. Differentiate $e^{\sin^{-1}x}$ w.r.t. x. [☞P 115(234)]

VSAQ 34. Find $\frac{dy}{dx}$ in $y=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ [☞P 116(235)]

VSAQ 35. Find $\frac{dy}{dx}$ of $x^2+xy+y^2=100$ [☞P 116(237)]

VSAQ 36. Find $\frac{dy}{dx}$, if $x=at^2, y=2at$. [☞P 116(238)]

VSAQ 37. If $y=A\sin x+B\cos x$, then prove that $\frac{d^2y}{dx^2}+y=0$ [☞P 117(243)]

★★★APPLICATION OF DERIVATIVES [2 MARKS] ★★★

VSAQ 38. Show that the function given by $f(x)=e^{2x}$ is increasing on R. [☞P 118(244)]

VSAQ 39. S.T the function f given by $f(x)=x^3-3x^2+4x, x\in R$ is increasing on R. [☞P 118(246)]

VSAQ 40. Find the intervals in which the function f given by $f(x)=x^2-4x+6$ is increasing [☞P 118(248)]

VSAQ 41. Find the maximum and minimum values of $f(x)=9x^2+12x+2$ [☞P 119(251)]

VSAQ 42. Find the rate of change of the area of a circle with respect to its radius r when r = 3 cm [☞P 119(253)]

★★★INTEGRATION [2 MARKS] ★★★

VSAQ 43. Find $\int \sec x(\sec x+\tan x)dx$ [☞P 120(258)] VSAQ 44. Find integral of $\frac{\sin x}{(1+\cos x)^2}$ [☞P 121(259)]

VSAQ 45. Find integral of $\frac{(1+\log x)^2}{x}$ [☞P 121(262)] VSAQ 46. Find integral of $\frac{e^{\tan^{-1}x}}{1+x^2}$ [☞P 121(260)]

VSAQ 47. Find integral of $\frac{(\log x)^2}{x}$ [☞P 121(262)]

VSAQ 48. Find integral of $e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)$ [☞P 122(264)] VSAQ 49. Evaluate $\int_2^3 \frac{x\,dx}{x^2+1}$ [☞P 123(272)]

VSAQ 50. Evaluate $\int_0^4 \frac{x^2}{1+x}dx$ [☞P 124(276)]

★★★DIFFERENTIAL EQUATION [2 MARKS] ★★★

VSAQ 51. Determine order and degree of $\left(\frac{ds}{dt}\right)^4+3s\frac{d^2s}{dt^2}=0$ [☞P 125(281)]

VSAQ 52. Determine order and degree of $y''+(y')^2+2y=0$ [☞P 125(282)]

VSAQ 53. Verify that the function $y=Ax$ is a solution to $xy'=y(x\neq 0)$ [☞P 126(284)]

VSAQ 54. Find the general solution of $\frac{dy}{dx}=(1+x^2)(1+y^2)$ [☞P 127(PQ)]

VSAQ 55. Find the general solution of $\frac{dy}{dx}+\frac{y}{x}=x^2$ [☞P 127(291)]

Mahesh

BBQ SPECIAL
To make you gain Confidence, Easy Scoring Chapters are brought to the beginning.

RISING STARS
MOST FAVOURITE SECTION

LAQ
SECTION-D

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LAQ CHAPTERS
5Q X 8 M= 40 MARKS

★★★ 1.	DETERMINANTS (DETS) (8 Marks)	16 - 18
★★★ 2.	VECTOR ALGEBRA (VA) (8 Marks)	19 - 21
★★★ 3.	3D GEOMETRY (3D) (8 Marks)	22 - 25
★★★ 4.	PROBABILITY (PROB) (8 Marks)	26 - 29
★★ 5.	APPLICATION OF DERIVATIVES(AD)(8 Marks)	30 - 34
★★★ 6.	INTEGRALS(IG) (8+8 Marks)	35 - 43
★★ 7.	DIFFERENTIAL EQUATIONS (DE)(8 Marks)	44 - 51
•	STAR QUESTIONS PLUS (SQP)	52

Mahesh

- The first 4 chapters are Easy, Important 'must & should' practice Chapters.
- In Integrals, **Definite Integral models** are few & important. Do them all.
- In Differential Equations, LDE Models are easy to do when compared to HDE.
- You should get prepared to attempt minimum 6 LAQs out of 8 LAQs in IPE.

Super Scoring Pages

★★★SSP★★★

Page 16, 17, 19 to 24, 26,27,28,35,36,37,44,45,46

Total SS Pages : 17

8 Marks

LAQ Prince Chapter

1 DETERMINANTS

LAQ First & Best Chapter

4 VERY IMP. QUESTIONS

1. **Solve the following system of equations by matrix method.**
$3x - 2y + 3z = 8,\ 2x + y - z = 1,\ 4x - 3y + 2z = 4$

Sol: Given system of equations is written in the form $AX = B$, where

$$A = \begin{bmatrix} 3 & -2 & 3 \\ 2 & 1 & -1 \\ 4 & -3 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix}$$

1 Mark

Now $|A| = 3(2-3) + 2(4+4) + 3(-6-4) = -3 + 16 - 30 = -17 \neq 0$

2 Marks

So A is non-singular. Hence A^{-1} exists.

Cofactors of matrix A:

$A_{11} = -1,\ A_{12} = -8,\ A_{13} = -10$

$A_{21} = -5,\ A_{22} = -6,\ A_{23} = 1$

$A_{31} = -1,\ A_{32} = 9,\ A_{33} = 7$

2 Marks

Hence $\text{adj}A = \begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 9 \\ -10 & 1 & 7 \end{bmatrix}$ $\therefore A^{-1} = \frac{1}{|A|}\text{adj}A = -\frac{1}{17}\begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 9 \\ -10 & 1 & 7 \end{bmatrix}$

1 Mark

Now $AX = B \Rightarrow X = A^{-1}B \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -\frac{1}{17}\begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 9 \\ -10 & 1 & 7 \end{bmatrix}\begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix} = -\frac{1}{17}\begin{bmatrix} -8-5-4 \\ -64-6+36 \\ -80+1+28 \end{bmatrix} = -\frac{1}{17}\begin{bmatrix} -17 \\ -34 \\ -51 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$\therefore$ the solution is $x = 1, y = 2, z = 3$.

2 Marks

☺ Hint Box ☺

$A_{11} = \begin{vmatrix} 1 & -1 \\ -3 & 2 \end{vmatrix} = 2 - 3 = -1$

$A_{12} = -\begin{vmatrix} 2 & -1 \\ 4 & 2 \end{vmatrix} = -(4+4) = -8$

$A_{13} = \begin{vmatrix} 2 & 1 \\ 4 & -3 \end{vmatrix} = -6 - 4 = -10$

☺ HINT BOX ☺

Hint 1: In the cofactor matrix +ve & -ve signs come alternatively

Hint 2: Except the center & 4 corner elements, remaining 4 have negative signs.

Hint 3: Adjoint A is the transpose of cofactor of matrix

Sign Pattern

$$\begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$$

In Q_1, the cofactor matrix of A:

$$\begin{bmatrix} \begin{vmatrix} 1 & -1 \\ -3 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & -1 \\ 4 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 4 & -3 \end{vmatrix} \\ -\begin{vmatrix} -2 & 3 \\ -3 & 2 \end{vmatrix} & \begin{vmatrix} 3 & 3 \\ 4 & 2 \end{vmatrix} & -\begin{vmatrix} 3 & -2 \\ 4 & -3 \end{vmatrix} \\ \begin{vmatrix} -2 & 3 \\ 1 & -1 \end{vmatrix} & -\begin{vmatrix} 3 & 3 \\ 2 & -1 \end{vmatrix} & \begin{vmatrix} 3 & -2 \\ 2 & 1 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} -1 & -8 & -10 \\ -5 & -6 & 1 \\ -1 & 9 & 7 \end{bmatrix}$$

Mahesh

2. **Solve the system of equations using matrix method**
$2x + 3y + 3z = 5,\ x - 2y + z = -4, 3x - y - 2z = 3$ **[BMP]**

Sol: Given system of equations is written in the form $AX = B$, where

$$A=\begin{bmatrix}2 & 3 & 3\\ 1 & -2 & 1\\ 3 & -1 & -2\end{bmatrix}, X=\begin{bmatrix}x\\ y\\ z\end{bmatrix} \text{ and } B=\begin{bmatrix}5\\ -4\\ 3\end{bmatrix}$$ **1 Mark**

Now $|A| = 2(4+1)-3(-2-3)+3(-1+6) = 10+15+15 = 40 \neq 0$ **2 Marks**

So A is non-singular. Hence A^{-1} exists.

Cofactors of matrix A:

$A_{11} = 5;\ A_{12} = 5;\ A_{13} = 5$

$A_{21} = 3;\ A_{22} = -13;\ A_{23} = 11$

$A_{31} = 9;\ A_{32} = 1;\ A_{33} = -7$ **2 Marks**

Hence $\text{adj}A=\begin{bmatrix}5 & 3 & 9\\ 5 & -13 & 1\\ 5 & 11 & -7\end{bmatrix}$ $\therefore A^{-1}=\frac{1}{|A|}\text{adj}A=\frac{1}{40}\begin{bmatrix}5 & 3 & 9\\ 5 & -13 & 1\\ 5 & 11 & -7\end{bmatrix}$ **1 Mark**

Now $AX=B \Rightarrow X=A^{-1}B \Rightarrow \begin{bmatrix}x\\ y\\ z\end{bmatrix}=\frac{1}{40}\begin{bmatrix}5 & 3 & 9\\ 5 & -13 & 1\\ 5 & 11 & -7\end{bmatrix}\begin{bmatrix}5\\ -4\\ 3\end{bmatrix}=\frac{1}{40}\begin{bmatrix}25-12+27\\ 25+52+3\\ 25-44-21\end{bmatrix}=\frac{1}{40}\begin{bmatrix}40\\ 80\\ -40\end{bmatrix}=\begin{bmatrix}1\\ 2\\ -1\end{bmatrix}$

$\therefore$ the solution is $x=1, y=2, z=-1$ **2 Marks**

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3. **Solve the system of equations using matrix method**
$x - y + 2z = 7,\ 3x + 4y - 5z = -5,\ 2x - y + 3z = 12$

Sol: Given system of equations is written in the form of $AX = B$ where

$$A=\begin{bmatrix}1 & -1 & 2\\ 3 & 4 & -5\\ 2 & -1 & 3\end{bmatrix}, X=\begin{bmatrix}x\\ y\\ z\end{bmatrix} \text{ and } B=\begin{bmatrix}7\\ -5\\ 12\end{bmatrix}$$ **1 Mark**

Now $|A| = 1(12-5)+1(9+10)+2(-3-8) = 7+19-22 = 4 \neq 0$ **2 Marks**

So A is non-singular. Hence, A^{-1} exists.

Cofactors of matrix A:

$A_{11} = 7;\ A_{12} = -19;\ A_{13} = -11$

$A_{21} = 1;\ A_{22} = -1;\ A_{23} = -1;$

$A_{31} = -3;\ A_{32} = 11;\ A_{33} = 7$ **2 Marks**

Hence $\text{adj}A=\begin{bmatrix}7 & 1 & -3\\ -19 & -1 & 11\\ -11 & -1 & 7\end{bmatrix}$ $\therefore A^{-1}=\frac{1}{|A|}\text{adj}A=\frac{1}{4}\begin{bmatrix}7 & 1 & -3\\ -19 & -1 & 11\\ -11 & -1 & 7\end{bmatrix}$ **1 Mark**

Now $AX=B \Rightarrow X=A^{-1}B \Rightarrow \begin{bmatrix}x\\ y\\ z\end{bmatrix}=\frac{1}{4}\begin{bmatrix}7 & 1 & -3\\ -19 & -1 & 11\\ -11 & -1 & 7\end{bmatrix}\begin{bmatrix}7\\ -5\\ 12\end{bmatrix}=\frac{1}{4}\begin{bmatrix}49-5-36\\ -133+5+132\\ -77+5+84\end{bmatrix}=\frac{1}{4}\begin{bmatrix}8\\ 4\\ 12\end{bmatrix}=\begin{bmatrix}2\\ 1\\ 3\end{bmatrix}$

2 Marks

$\therefore$ the solution is $x=2, y=1, z=3$

Manish

4. If $A=\begin{bmatrix}2 & -3 & 5\\ 3 & 2 & -4\\ 1 & 1 & -2\end{bmatrix}$, find A^{-1}. Using A^{-1} solve the system of equations

$2x-3y+5z=11,\ 3x+2y-4z=-5,\ x+y-2z=-3$

Sol: Given system of equations is written in the form of $AX=B$ where

$$A=\begin{bmatrix}2 & -3 & 5\\ 3 & 2 & -4\\ 1 & 1 & -2\end{bmatrix}, X=\begin{bmatrix}x\\ y\\ z\end{bmatrix} \text{ and } B=\begin{bmatrix}11\\ -5\\ -3\end{bmatrix}$$ **1 Mark**

Now $|A|=2(-4+4)+3(-6+4)+5(3-2)=0-6+5=-1\neq 0$ **2 Marks**

So A is non-singular. Hence, A^{-1} exists.

Cofactors of matrix A:

$A_{11}=0;\ A_{12}=2;\ A_{13}=1;$

$A_{21}=-1;\ A_{22}=-9;\ A_{23}=-5$

$A_{31}=2;\ A_{32}=23;\ A_{33}=13$ **2 Marks**

Hence $\text{adj}A=\begin{bmatrix}0 & -1 & 2\\ 2 & -9 & 23\\ 1 & -5 & 13\end{bmatrix}$

$$\therefore A^{-1}=\frac{1}{|A|}\text{adj}A=-\begin{bmatrix}0 & -1 & 2\\ 2 & -9 & 23\\ 1 & -5 & 13\end{bmatrix}=\begin{bmatrix}0 & 1 & -2\\ -2 & 9 & -23\\ -1 & 5 & -13\end{bmatrix}$$ **1 Mark**

$$\text{Now } AX=B\Rightarrow X=A^{-1}B\Rightarrow\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}0 & 1 & -2\\ -2 & 9 & -23\\ -1 & 5 & -13\end{bmatrix}\begin{bmatrix}11\\ -5\\ -3\end{bmatrix}=\begin{bmatrix}0-5+6\\ -22-45+69\\ -11-25+39\end{bmatrix}=\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}$$

$\therefore$ the solution is $x=1, y=2$ and $z=3$. **2 Marks**

PQ **Solve the system of equations using matrix method**
$x-y+z=4,\ 2x+y-3z=0,\ x+y+z=2$

Hint 1: $|A|=10$

Hint 2: $\text{adj}A=\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & 5\\ 1 & -2 & 3\end{bmatrix}; A^{-1}=\frac{1}{10}\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & 5\\ 1 & -2 & 3\end{bmatrix}$

Hint 3: Solution: $x=2, y=-1, z=1$

[For Answer ref. QR Code on P.No.52]

Mahesh

'Must& Should' Practice Chapter

8 Marks

2 VECTOR ALGEBRA

6 EASY & SURE SHOT QUESTIONS

LAQ King Chapter

5. If $\bar{a}=2\bar{i}+3\bar{j}+4\bar{k}, \bar{b}=\bar{i}+\bar{j}-\bar{k}, \bar{c}=\bar{i}-\bar{j}+\bar{k}$, compute $\bar{a}\times(\bar{b}\times\bar{c})$ and verify that it is perpendicular to $\bar{a}$ **[BMP]**

Sol: Given $\bar{a}=2\bar{i}+3\bar{j}+4\bar{k}, \bar{b}=\bar{i}+\bar{j}-\bar{k}, \bar{c}=\bar{i}-\bar{j}+\bar{k}$

To find $\bar{a}\times(\bar{b}\times\bar{c})$, first we have to find the bracket term $(\bar{b}\times\bar{c})$

(i) $\bar{b}\times\bar{c}=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\1&1&-1\\1&-1&1\end{vmatrix}=\bar{i}(1-1)-\bar{j}(1+1)+\bar{k}(-1-1)=-2\bar{j}-2\bar{k}$ **3 Marks**

(ii) $\bar{a}\times(\bar{b}\times\bar{c})=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\2&3&4\\0&-2&-2\end{vmatrix}=\bar{i}(-6+8)-\bar{j}(-4-0)+\bar{k}(-4-0)=2\bar{i}+4\bar{j}-4\bar{k}$ **3 Marks**

(iii) $[\bar{a}\times(\bar{b}\times\bar{c}))].\bar{a}=(2\bar{i}+4\bar{j}-4\bar{k}).(2\bar{i}+3\bar{j}+4\bar{k})=4+12-16=0$ **1 Mark**

(iv) $\bar{a}\times(\bar{b}\times\bar{c})$ is perpendicular to $\bar{a}$. [$\because$ Dot product of perpendicular vectors is 0] **1 Mark**

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6. If $\vec{a}=2\vec{i}+\vec{j}-3\vec{k}$, $\vec{b}=\vec{i}-2\vec{j}+\vec{k}$, $\vec{c}=-\vec{i}+\vec{j}-4\vec{k}$ and $\vec{d}=\vec{i}+\vec{j}+\vec{k}$ then compute $|(\vec{a}\times\vec{b})\times(\vec{c}+\vec{d})|$

Sol: (i) Given vectors $\vec{a}=2\vec{i}+\vec{j}-3\vec{k}$ and $\vec{b}=\vec{i}-2\vec{j}+\vec{k}$

$$\vec{a}\times\vec{b}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\2&1&-3\\1&-2&1\end{vmatrix}=\vec{i}(1-6)-\vec{j}(2+3)+\vec{k}(-4-1)=-5\vec{i}-5\vec{j}-5\vec{k}$$ **2 Marks**

(ii) Given $\vec{c}=-\vec{i}+\vec{j}-4\vec{k}$ and $\vec{d}=\vec{i}+\vec{j}+\vec{k}$

$\Rightarrow \vec{c}+\vec{d}=(-\vec{i}+\vec{j}-4\vec{k})+(\vec{i}+\vec{j}+\vec{k})=0\hat{i}+2\hat{j}-3\hat{k}$

Now $\vec{a}\times\vec{b}=-5\hat{i}-5\hat{j}-5\hat{k}$ and $\vec{c}+\vec{d}=0\hat{i}+2\hat{j}-3\hat{k}$, then **2 Marks**

(iii) $(\vec{a}\times\vec{b})\times(\vec{c}+\vec{d})=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-5&-5&-5\\0&2&-3\end{vmatrix}=\hat{i}(15+10)-\hat{j}(15-0)+\hat{k}(-10-0)$

$=25\hat{i}-15\hat{j}-10\hat{k}=5(5\hat{i}-3\hat{j}-2\hat{k})$ **2 Marks**

(iv) $\therefore|(\vec{a}\times\vec{b})\times(\vec{c}+\vec{d})|=5|5\hat{i}-3\hat{j}-2\hat{k}|=5\sqrt{5^2+(-3)^2+(-2)^2}=5\sqrt{25+9+4}=5\sqrt{38}$ **2 Marks**

Mahesh

7. If $\bar{a}=\bar{i}-2\bar{j}-3\bar{k}$, $\bar{b}=2\bar{i}+\bar{j}-\bar{k}, \bar{c}=\bar{i}+3\bar{j}-2\bar{k}$ then, verify $\bar{a}\times(\bar{b}\times\bar{c})\neq(\bar{a}\times\bar{b})\times\bar{c}$

Sol: **1) To find $\bar{a}\times(\bar{b}\times\bar{c})$, first we have to find bracket term $(\bar{b}\times\bar{c})$**

(i) $\bar{b}\times\bar{c}=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\2&1&-1\\1&3&-2\end{vmatrix}=\bar{i}(-2+3)-\bar{j}(-4+1)+\bar{k}(6-1)=\bar{i}+3\bar{j}+5\bar{k}$ **2 Marks**

(ii) $\bar{a}\times(\bar{b}\times\bar{c})=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\1&-2&-3\\1&3&5\end{vmatrix}=\bar{i}(-10+9)-\bar{j}(5+3)+\bar{k}(3+2)=-\bar{i}-8\bar{j}+5\bar{k}$ ....(1) **2 Marks**

2) To find $(\bar{a}\times\bar{b})\times\bar{c}$ we have to find bracket term $(\bar{a}\times\bar{b})$

(iii) $\bar{a}\times\bar{b}=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\1&-2&-3\\2&1&-1\end{vmatrix}=\bar{i}(2+3)-\bar{j}(-1+6)+\bar{k}(1+4)=5\bar{i}-5\bar{j}+5\bar{k}$ **2 Marks**

(iv) $(\bar{a}\times\bar{b})\times\bar{c}=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\5&-5&5\\1&3&-2\end{vmatrix}=\bar{i}(10-5)-\bar{j}(-10-5)+\bar{k}(15+5)=-5\bar{i}+15\bar{j}+20\bar{k}$ ....(2) **2 Marks**

$\therefore$ From (1) & (2), $\bar{a}\times(\bar{b}\times\bar{c})\neq(\bar{a}\times\bar{b})\times\bar{c}$

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8. If $\bar{a}=7\bar{i}-2\bar{j}+3\bar{k}, \bar{b}=2\bar{i}+8\bar{k}$ and $\bar{c}=\bar{i}+\bar{j}+\bar{k}$, then compute $\bar{a}\times\bar{b}, \bar{a}\times\bar{c}$ and $\bar{a}\times(\bar{b}+\bar{c})$.

Verify whether cross product is distributive over vector addition.

Sol: Given that $\bar{a}=7\bar{i}-2\bar{j}+\bar{k}, \bar{b}=2\bar{i}+8\bar{k}$ and $\bar{c}=\bar{i}+\bar{j}+\bar{k}$

(i) $\bar{a}\times\bar{b}=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\7&-2&3\\2&0&8\end{vmatrix}=\bar{i}(-16-0)-\bar{j}(56-6)+\bar{k}(0+4)=-16\bar{i}-50\bar{j}+4\bar{k}$ **2 Marks**

(ii) $\bar{a}\times\bar{c}=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\7&-2&3\\1&1&1\end{vmatrix}=\bar{i}(-2-3)-\bar{j}(7-3)+\bar{k}(7+2)=-5\bar{i}-4\bar{j}+9\bar{k}$

Now $\bar{b}+\bar{c}=(2\bar{i}+8\bar{k})+(\bar{i}+\bar{j}+\bar{k})=3\bar{i}+\bar{j}+9\bar{k}$ **2 Marks**

(iii) $\bar{a}\times(\bar{b}+\bar{c})=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\7&-2&3\\3&1&9\end{vmatrix}=\bar{i}(-18-3)-\bar{j}(63-9)+\bar{k}(7+6)=-21\bar{i}-54\bar{j}+13\bar{k}$.....(1) **2 Marks**

(iv) Now $(\bar{a}\times\bar{b})+(\bar{a}\times\bar{c})=(-16\bar{i}-50\bar{j}+4\bar{k})+(-5\bar{i}-4\bar{j}+9\bar{k})=-21\bar{i}-54\bar{j}+13\bar{k}$.....(2)

From (1) & (2), we have $\bar{a}\times(\bar{b}+\bar{c})=(\bar{a}\times\bar{b})+(\bar{a}\times\bar{c})$

$\therefore$ Vector product is distributive over vector addition. **2 Marks**

Mahesh

9. If $\bar{a}=\bar{i}-2\bar{j}+3\bar{k}, \bar{b}=2\bar{i}+\bar{j}+\bar{k}, \bar{c}=\bar{i}+\bar{j}+2\bar{k}$ then find $|(\bar{a}\times\bar{b})\times\bar{c}|$ and $|\bar{a}\times(\bar{b}\times\bar{c})|$

Sol: Given $\bar{a}=\bar{i}-2\bar{j}+3\bar{k}, \bar{b}=2\bar{i}+\bar{j}+\bar{k}, \bar{c}=\bar{i}+\bar{j}+2\bar{k}$

1) To find $(\bar{a}\times\bar{b})\times\bar{c}$, first we have to find bracket term $\bar{a}\times\bar{b}$

(i) $$\bar{a}\times\bar{b}=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\1&-2&3\\2&1&1\end{vmatrix}=\bar{i}(-2-3)-\bar{j}(1-6)+\bar{k}(1+4)=-5\bar{i}+5\bar{j}+5\bar{k}$$ **2 Marks**

(ii) $$(\bar{a}\times\bar{b})\times\bar{c}=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\-5&5&5\\1&1&2\end{vmatrix}=\bar{i}(10-5)-\bar{j}(-10-5)+\bar{k}(-5-5)$$

$$=5\bar{i}+15\bar{j}-10\bar{k}=5(\bar{i}+3\bar{j}-2\bar{k})$$

$$\therefore|(a\times b)\times c|=5|\bar{i}+3\bar{j}-2\bar{k}|=5\sqrt{1^2+3^2+(-2)^2}=5\sqrt{1+9+4}=5\sqrt{14}$$ **2 Marks**

2) To find $\bar{a}\times(\bar{b}\times\bar{c})$ we have to find bracket term $\bar{b}\times\bar{c}$

(iii) $$\bar{b}\times\bar{c}=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\2&1&1\\1&1&2\end{vmatrix}=\bar{i}(2-1)-\bar{j}(4-1)+\bar{k}(2-1)=\bar{i}-3\bar{j}+\bar{k}$$ **2 Marks**

(iv) $$\bar{a}\times(\bar{b}\times\bar{c})=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\1&-2&3\\1&-3&1\end{vmatrix}=\bar{i}(-2+9)-\bar{j}(1-3)+\bar{k}(-3+2)=7\bar{i}+2\bar{j}-\bar{k}$$

$$\therefore|\bar{a}\times(\bar{b}\times\bar{c})|=|7\bar{i}+2\bar{j}-\bar{k}|=\sqrt{7^2+2^2+(-1)^2}=\sqrt{49+4+1}=\sqrt{54}=\sqrt{9\times6}=3\sqrt{6}$$ **2 Marks**

---

10. **Find the area of the triangle with vertices A(1, 1, 2), B(2, 3, 5) and C(1, 5, 5).**

Sol: (i) Vertices of Δ ABC are A(1, 1, 2), B(2, 3, 5), C(1, 5, 5).

Then we have $\overrightarrow{OA}=\hat{i}+\hat{j}+2\hat{k}$, $\overrightarrow{OB}=2\hat{i}+3\hat{j}+5\hat{k}$, $\overrightarrow{OC}=\hat{i}+5\hat{j}+5\hat{k}$ **2 Marks**

(ii) $\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=(2\hat{i}+3\hat{j}+5\hat{k})-(\hat{i}+\hat{j}+2\hat{k})=\hat{i}+2\hat{j}+3\hat{k}$

$\overrightarrow{AC}=\overrightarrow{OC}-\overrightarrow{OA}=(\hat{i}+5\hat{j}+5\hat{k})-(\hat{i}+\hat{j}+2\hat{k})=0\hat{i}+4\hat{j}+3\hat{k}$ **2 Marks**

(iii) $$\overrightarrow{AB}\times\overrightarrow{AC}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&2&3\\0&4&3\end{vmatrix}=\hat{i}(6-12)-\hat{j}(3-0)+\hat{k}(4-0)=-6\hat{i}-3\hat{j}+4\hat{k}$$ **2 Marks**

(iv) Area of Δ ABC $=\frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|=\frac{1}{2}\sqrt{(-6)^2+(-3)^2+4^2}=\frac{1}{2}\sqrt{36+9+16}=\frac{\sqrt{61}}{2}$ Sq.units. **2 Marks**

Mahesh

LAQ Queen Chapter

3

8 Marks

3D GEOMETRY

TWO IMP. MODELS

Sure Shot Shortest Chapter

11. **Find the shortest distance between the lines** [BMP]

$\vec{r}=(\hat{i}+2\hat{j}+\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=2\hat{i}-\hat{j}-\hat{k}+\mu(2\hat{i}+\hat{j}+2\hat{k})$

Sol: Comparing the given skew lines with $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$, we get

$\vec{a_1}=(\hat{i}+2\hat{j}+\hat{k})$, $\vec{b_1}=(\hat{i}-\hat{j}+\hat{k})$ and $\vec{a}_2=(2\hat{i}-\hat{j}-\hat{k})$, $\vec{b}_2=(2\hat{i}+\hat{j}+2\hat{k})$ 1 Mark

Now $\vec{a}_1-\vec{a}_2=(\hat{i}+2\hat{j}+\hat{k})-(2\hat{i}-\hat{j}-\hat{k})=-\hat{i}+3\hat{j}+2\hat{k}$ 1 Mark

$$\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&1\\2&1&2\end{vmatrix}=(-2-1)\hat{i}-(2-2)\hat{j}+(1+2)\hat{k}=-3\hat{i}+3\hat{k}$$ 2 Marks

$$|\vec{b_1}\times\vec{b_2}|=\sqrt{(-3)^2+3^2}=\sqrt{9+9}=\sqrt{18}=\sqrt{9\times2}=3\sqrt{2}$$ 1 Mark

Shortest distance between the lines is $d=\dfrac{|(\vec{a_1}-\vec{a_2}).(\vec{b_1}\times\vec{b_2})|}{|\vec{b}_1\times\vec{b}_2|}$ 1 Mark

$$d=\frac{|(-\hat{i}+3\hat{j}+2\hat{k}).(-3\hat{i}+3\hat{k})|}{3\sqrt{2}}=\frac{|(-1)(-3)+3(0)+2(3)|}{3\sqrt{2}}=\frac{|9|}{3\sqrt{2}}=\frac{3}{\sqrt{2}}=\frac{3}{\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}}=\frac{3\sqrt{2}}{2}$$ 2 Marks

12. **Find the shortest distance between the lines whose vector equations are**

$\vec{r}=(\hat{i}+2\hat{j}+3\hat{k})+\lambda(\hat{i}-3\hat{j}+2\hat{k})$ and $\vec{r}=4\hat{i}+5\hat{j}+6\hat{k}+\mu(2\hat{i}+3\hat{j}+\hat{k})$

Sol: Comparing the given skew lines with $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$ we get

$\vec{a_1}=(\hat{i}+2\hat{j}+3\hat{k})$, $\vec{b_1}=(\hat{i}-3\hat{j}+2\hat{k})$ and $\vec{a}_2=(4\hat{i}+5\hat{j}+6\hat{k})$, $\vec{b}_2=(2\hat{i}+3\hat{j}+\hat{k})$ 1 Mark

Now $\vec{a_1}-\vec{a_2}=(\hat{i}+2\hat{j}+3\hat{k})-(4\hat{i}+5\hat{j}+6\hat{k})=-3\hat{i}-3\hat{j}-3\hat{k}$ 1 Mark

$$\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-3&2\\2&3&1\end{vmatrix}=(-3-6)\hat{i}-(1-4)\hat{j}+(3+6)\hat{k}=-9\hat{i}+3\hat{j}+9\hat{k}$$ 2 Marks

$$|\vec{b_1}\times\vec{b_2}|=\sqrt{(-9)^2+(3)^2+(9)^2}=\sqrt{81+9+81}=\sqrt{171}=\sqrt{9\times19}=3\sqrt{19}$$ 1 Mark

Shortest distance between the lines is $d=\dfrac{|(\vec{a_1}-\vec{a_2}).(\vec{b_1}\times\vec{b_2})|}{|\vec{b}_1\times\vec{b}_2|}$ 1 Mark

$$d=\frac{|(-3\hat{i}-3\hat{j}-3\hat{k}).(-9\hat{i}+3\hat{j}+9\hat{k})|}{3\sqrt{19}}=\frac{|(-3)(-9)-3\times3-3\times9|}{3\sqrt{19}}=\frac{|27-9-27|}{3\sqrt{19}}=\frac{|-9|}{3\sqrt{19}}=\frac{\cancel{9}}{\cancel{3}\sqrt{19}}=\frac{3}{\sqrt{19}}$$

2 Marks

Mahesh

13. **Find the shortest distance between the lines l_1 and l_2 whose vector equations are $\vec{r}=(\hat{i}+\hat{j})+\lambda(2\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=(2\hat{i}+\hat{j}-\hat{k})+\mu(3\hat{i}-5\hat{j}+2\hat{k})$**

Sol: Comparing the given skew lines with $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$ we get

$\vec{a}_1=\hat{i}+\hat{j}, \vec{b}_1=2\hat{i}-\hat{j}+\hat{k}$ and $\vec{a}_2=2\hat{i}+\hat{j}-\hat{k}$, $\vec{b}_2=3\hat{i}-5\hat{j}+2\hat{k}$

Here $\vec{a}_1-\vec{a}_2=(\hat{i}+\hat{j})-(2\hat{i}+\hat{j}-\hat{k})=-\hat{i}+\hat{k}$ — 1 Mark

1 Mark

$$\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&-1&1\\3&-5&2\end{vmatrix}=(-2+5)\hat{i}-(4-3)\hat{j}+(-10+3)\hat{k}=3\hat{i}-\hat{j}-7\hat{k}$$ — 2 Marks

$$|\vec{b}_1\times\vec{b}_2|=\sqrt{3^2+(-1)^2+(-7)^2}=\sqrt{9+1+49}=\sqrt{59}$$ — 1 Mark

Shortest distance between the lines is $d=\dfrac{|(\vec{a_1}-\vec{a_2}).(\vec{b_1}\times\vec{b_2})|}{|\vec{b_1}\times\vec{b_2}|}$ — 1 Mark

$$d=\frac{|(-\hat{i}+\hat{k}).(3\hat{i}-\hat{j}-7\hat{k})|}{\sqrt{59}}=\frac{|-3+0-7|}{\sqrt{59}}=\frac{|-10|}{\sqrt{59}}=\frac{10}{\sqrt{59}}$$ — 2 Marks

PQ **Find the shortest distance between lines** **[For Answer ref. QR Code on P.No.52]**

$\vec{r}=6\hat{i}+2\hat{j}+2\hat{k}+\lambda(\hat{i}-2\hat{j}+2\hat{k})$ **and** $\vec{r}=-4\hat{i}-\hat{k}+\mu(3\hat{i}-2\hat{j}-2\hat{k})$ **[Ans: 9 units]**

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14. **Find the shortest distance between the lines whose vector equations are $\vec{r}=(1-t)\hat{i}+(t-2)\hat{j}+(3-2t)\hat{k}$ and $\vec{r}=(s+1)\hat{i}+(2s-1)\hat{j}-(2s+1)\hat{k}$**

Sol: First line is $\vec{r}=(1-t)\hat{i}+(t-2)\hat{j}+(3-2t)\hat{k}=(\hat{i}-2\hat{j}+3\hat{k})+t(-\hat{i}+\hat{j}-2\hat{k})$

Second line is $\vec{r}=(s+1)\hat{i}+(2s-1)\hat{j}-(2s+1)\hat{k}=(\hat{i}-\hat{j}-\hat{k})+s(\hat{i}+2\hat{j}-2\hat{k})$

Comparing the given skew lines with $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$ we get

$\vec{a_1}=(\hat{i}-2\hat{j}+3\hat{k})$, $\vec{b}_1=(-\hat{i}+\hat{j}-2\hat{k})$ and $\vec{a_2}=(\hat{i}-\hat{j}-\hat{k})$, $\vec{b}_2=(\hat{i}+2\hat{j}-2\hat{k})$ — 1 Mark

$\vec{a_1}-\vec{a_2}=(\hat{i}-2\hat{j}+3\hat{k})-(\hat{i}-\hat{j}-\hat{k})=-\hat{j}+4\hat{k}$ — 1 Mark

$$\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-1&1&-2\\1&2&-2\end{vmatrix}=(-2+4)\hat{i}-(2+2)\hat{j}+(-2-1)\hat{k}=2\hat{i}-4\hat{j}-3\hat{k}$$ — 2 Marks

$$|\vec{b_1}\times\vec{b_2}|=\sqrt{(2)^2+(-4)^2+(-3)^2}=\sqrt{4+16+9}=\sqrt{29}$$ — 1 Mark

Shortest distance between the lines is $d=\dfrac{|(\vec{a_1}-\vec{a_2}).(\vec{b_1}\times\vec{b_2})|}{|\vec{b_1}\times\vec{b_2}|}$ — 1 Mark

$$d=\frac{|(-\hat{j}+4\hat{k}).(2\hat{i}-4\hat{j}-3\hat{k})|}{\sqrt{29}}=\frac{|0+(-1)(-4)+(4)(-3)|}{\sqrt{29}}=\frac{|4-12|}{\sqrt{29}}=\frac{|-8|}{\sqrt{29}}=\frac{8}{\sqrt{29}}\text{ units.}$$ — 2 Marks

Mahesh

15. Find the shortest distance between the lines

$\frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1}$ and $\frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1}$

Sol: Reducing the given skew lines in the vector form, we get

$\vec{a}_1=-\hat{i}-\hat{j}-\hat{k}, \vec{b}_1=7\hat{i}-6\hat{j}+\hat{k}$ and $\vec{a}_2=3\hat{i}+5\hat{j}+7\hat{k}$, $\vec{b}_2=\hat{i}-2\hat{j}+\hat{k}$ **1 Mark**

Now $\vec{a}_1-\vec{a}_2=(-\hat{i}-\hat{j}-\hat{k})-(3\hat{i}+5\hat{j}+7\hat{k})=-4\hat{i}-6\hat{j}-8\hat{k}$ **1 Mark**

$$\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\7&-6&1\\1&-2&1\end{vmatrix}=\hat{i}(-6+2)-\hat{j}(7-1)+\hat{k}(-14+6)=-4\hat{i}-6\hat{j}-8\hat{k}$$ **2 Marks**

$|\vec{b}_1\times\vec{b}_2|=\sqrt{16+36+64}=\sqrt{116}$ **1 Mark**

Shortest distance between the lines is $d=\frac{|(\vec{a}_1-\vec{a}_2).(\vec{b}_1\times\vec{b}_2)|}{|\vec{b}_1\times\vec{b}_2|}$ **1 Mark**

$$d=\frac{|(-4\hat{i}-6\hat{j}-8\hat{k}).(-4\hat{i}-6\hat{j}-8\hat{k})|}{\sqrt{116}}=\frac{|16+36+64|}{\sqrt{116}}=\frac{116}{\sqrt{116}}=\sqrt{116}=\sqrt{4\times29}=2\sqrt{29}$$ **2 Marks**

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16. Find the distance between the pair of parallel lines

$\vec{r}=(\hat{i}+2\hat{j}-4\hat{k})+\lambda(4\hat{i}+3\hat{j}-12\hat{k})$ and $\vec{r}=(3\hat{i}+3\hat{j}-5\hat{k})+\mu(4\hat{i}+3\hat{j}-12\hat{k})$

Sol: Comparing the given parallel lines with $\vec{r}=\vec{a}_1+\lambda\vec{b}$ and $\vec{r}=\vec{a}_2+\mu\vec{b}$ we get

$\vec{a}_1=\hat{i}+2\hat{j}-4\hat{k}$, $\vec{a}_2=3\hat{i}+3\hat{j}-4\hat{k}$ and $\vec{b}=4\hat{i}+3\hat{j}-12\hat{k}$ **1 Mark**

Now $\vec{a}_1-\vec{a}_2=(\hat{i}+2\hat{j}-4\hat{k})-(3\hat{i}+3\hat{j}-4\hat{k})=-2\hat{i}-\hat{j}+\hat{k}$ **1 Mark**

Also $\vec{b}=4\hat{i}+3\hat{j}-12\hat{k}\Rightarrow|\vec{b}|=\sqrt{4^2+3^2+(-12)^2}=\sqrt{16+9+144}=\sqrt{169}=13$ **1 Mark**

$$\text{Now } (\vec{a}_1-\vec{a}_2)\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-2&-1&1\\4&3&-12\end{vmatrix}=(12-3)\hat{i}-(24-4)\hat{j}+(-6+4)\hat{k}=9\hat{i}-20\hat{j}-2\hat{k}$$ **2 Marks**

$|(\vec{a}_1-\vec{a}_2)\times\vec{b}|=\sqrt{9^2+(-20)^2+(-2)^2}=\sqrt{81+400+4}=\sqrt{485}$ **1 Mark**

$\therefore$ Distance between given parallel lines $d=\frac{|(\vec{a}_1-\vec{a}_2)\times\vec{b}|}{|\vec{b}|}=\frac{\sqrt{485}}{13}$ **2 Marks**

Q **Find the distance between the lines l_1 and l_2 given by**

$\vec{r}=\hat{i}+2\hat{j}-4\hat{k}+\lambda(2\hat{i}+3\hat{j}+6\hat{k})$, $\vec{r}=3\hat{i}+3\hat{j}-5\hat{k}+\mu(2\hat{i}+3\hat{j}+6\hat{k})$ $\left[\text{Ans}:\frac{\sqrt{293}}{7}\right]$

Mahesh

17. **Find the distance between the pair of parallel llines**

$\vec{r}=(\hat{i}+7\hat{j}+4\hat{k})+\lambda(3\hat{i}+2\hat{j}+5\hat{k})$ **and** $\vec{r}=(3\hat{i}+6\hat{j}+4\hat{k})+\mu(3\hat{i}+2\hat{j}+5\hat{k})$

Sol: Comparing the given parallel lines with $\vec{r}=\vec{a}_1+\lambda\vec{b}$ and $\vec{r}=\vec{a}_2+\mu\vec{b}$, we get

$\vec{a}_1=\hat{i}+7\hat{j}+4\hat{k}$, $\vec{a}_2=3\hat{i}+6\hat{j}+4\hat{k}$ and $\vec{b}=3\hat{i}+2\hat{j}+5\hat{k}$ **1 Mark**

Now $\vec{a}_1-\vec{a}_2=(\hat{i}+7\hat{j}+4\hat{k})-(3\hat{i}+6\hat{j}+4\hat{k})=-2\hat{i}+\hat{j}+0\hat{k}$ **1 Mark**

Also $\vec{b}=3\hat{i}+2\hat{j}+5\hat{k}\Rightarrow|\vec{b}|=\sqrt{3^2+2^2+5^2}=\sqrt{9+4+25}=\sqrt{38}$ **1 Mark**

Now $(\vec{a}_1-\vec{a}_2)\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-2&1&0\\3&2&5\end{vmatrix}=(5-0)\hat{i}-(-10-0)\hat{j}+(-4-3)\hat{k}=5\hat{i}+10\hat{j}-7\hat{k}$ **2 Marks**

$|(\vec{a}_1-\vec{a}_2)\times\vec{b}|=\sqrt{5^2+10^2+(-7)^2}=\sqrt{25+100+49}=\sqrt{174}$ **1 Mark**

$\therefore$ Distance between given parallel lines $d=\dfrac{|(\vec{a}_1-\vec{a}_2)\times\vec{b}|}{|\vec{b}|}=\dfrac{\sqrt{174}}{\sqrt{38}}=\dfrac{\sqrt{2\times87}}{\sqrt{2\times19}}=\sqrt{\dfrac{87}{19}}$ **2 Marks**

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18. **Find the distance between the pair of parallel lines**

$\dfrac{x+3}{-2}=\dfrac{y-5}{3}=\dfrac{z+2}{-1}$ and $\dfrac{x+2}{2}=\dfrac{y+2}{-3}=\dfrac{z-3}{1}$

Sol: Reducing the given parallel lines in the vector form, we get

Hence $\vec{a}_1=(-3,5,-2)=-3\hat{i}+5\hat{j}-2\hat{k}$; $\vec{a}_2=(-2,-2,3)=-2\hat{i}-2\hat{j}+3\hat{k}$ **1 Mark**

$\vec{b}=(-2,3,-1)=-2\hat{i}+3\hat{j}-\hat{k}\Rightarrow|\vec{b}|=\sqrt{(-2)^2+3^2+(-1)^2}=\sqrt{4+9+1}=\sqrt{14}$ **1 Mark**

$\Rightarrow\vec{a}_1-\vec{a}_2=(-3\hat{i}+5\hat{j}-2\hat{k})-(-2\hat{i}-2\hat{j}+3\hat{k})=-\hat{i}+7\hat{j}-5\hat{k}$ **1 Mark**

Now $(\vec{a}_1-\vec{a}_2)\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-1&7&-5\\-2&3&-1\end{vmatrix}=(-7+15)\hat{i}-(1+10)\hat{j}+(-3+14)\hat{k}=8\hat{i}+9\hat{j}+11\hat{k}$ **2 Marks**

$|(\vec{a}_1-\vec{a}_2)\times\vec{b}|=\sqrt{8^2+9^2+11^2}=\sqrt{64+81+121}=\sqrt{266}$ **1 Mark**

Distance between given parallel lines $d=\dfrac{|(\vec{a}_1-\vec{a}_2)\times\vec{b}|}{|\vec{b}|}=\sqrt{\dfrac{266}{14}}=\sqrt{\dfrac{2\times133}{2\times7}}=\sqrt{\dfrac{133}{7}}$ **2 Marks**

PQ **Find the distance between the pair of parallel lines**

$\dfrac{x-1}{1}=\dfrac{y-2}{2}=\dfrac{z-3}{-3}$ and $\dfrac{x+2}{-1}=\dfrac{y+1}{-2}=\dfrac{z+2}{3}$ $\left[\text{Ans}:\sqrt{\dfrac{283}{7}}\right]$

Mahesh

8 Marks

LAQ Princess Chapter

4 PROBABILITY

Probably The Best Chapter

5 BEST BAYE'S QUESTIONS

19. **Bag I contains 3 red and 4 black balls while another Bag II contains 5 red and 6 black balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from Bag II.**

Sol: Let E_1, E_2 be the events of selecting Bag I, Bag II respectively.

Then $P(E_1) = P(E_2) = \frac{1}{2}$ **2 Marks**

B-I	B-II
3 R	5 R
4 B	6 B

Let A be the event of drawing a red ball, then

$P(A|E_1)$ = P(drawing a red ball from Bag I) $= \frac{3}{7}$

$P(A|E_2)$ = P(drawing a red ball from Bag II) $= \frac{5}{11}$ **2 Marks**

$\therefore P(E_2|A)$ is obtained by Baye's theorem.

$$P(E_2|A) = \frac{P(E_2)P(A|E_2)}{P(E_1)P(A|E_1)+P(E_2)P(A|E_2)} = \frac{\frac{1}{2}\times\frac{5}{11}}{\left(\frac{1}{2}\times\frac{3}{7}\right)+\left(\frac{1}{2}\times\frac{5}{11}\right)} = \frac{\frac{5}{11}}{\frac{3}{7}+\frac{5}{11}} = \frac{\frac{5}{11}}{\frac{33+35}{77}} = \frac{5}{11}\times\frac{77}{68} = \frac{35}{68}$$

4 Marks

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20. **A bag contains 4 red and 4 black balls, another bag contains 2 red and 6 black balls. One of the two bags is selected at random and a ball is drawn from the bag which is found to be red. Find the probability that the ball is drawn from the first bag.**

Sol: Let E_1, E_2 be the events of selecting Bag I, Bag II respectively.

Then $P(E_1) = P(E_2) = \frac{1}{2}$ **2 Marks**

B-I	B-II
4 R	2 R
4 B	6 B

Let A be the event of getting a red ball, then

$P(A|E_1)$=P (drawing a red ball from first bag) $= \frac{4}{8} = \frac{1}{2}$

$P(A|E_2)$=P (drawing a red ball from second bag) $= \frac{2}{8} = \frac{1}{4}$ **2 Marks**

$\therefore P(E_1|A)$ is obtained by Baye's theorem.

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1)+P(E_2)P(A|E_2)} = \frac{\frac{1}{2}\times\frac{1}{2}}{\left(\frac{1}{2}\times\frac{1}{2}\right)+\left(\frac{1}{2}\times\frac{1}{4}\right)} = \frac{\frac{1}{4}}{\frac{1}{4}+\frac{1}{8}} = \frac{\frac{1}{4}}{\frac{3}{8}} = \frac{2}{3}$$

4 Marks

PQ **Bag I contains 3 red and 4 black balls and Bag II contains 4 red and 5 black balls. One ball is transferred from Bag I to Bag II and then a ball is drawn from Bag II. The ball so drawn is found to be red in colour. Find the probability that the transferred ball is black.**

[Ans: 16/31]

Manesh

21. **Given three identical boxes I, II and III, each containing two coins. In box I, both coins are gold coins, in box II, both are silver coins and in the box III, there is one gold and one silver coin. A person chooses a box at random and takes out a coin. If the coin is of gold, what is the probability that the other coin in the box is also of gold?**

Sol: Let E_1, E_2 , E_3 be the events of selecting boxes I, II , III respectively.

Then $P(E_1) = P(E_2) = P(E_3)=1/3$

Let A be the event of drawing a gold coin. **2 Marks**

B-I	B-II	B-III
2G	2S	1G, 1S

$P(A|E_1)$ = P(gold coin from bag I) $=\frac{2}{2}=1$

$P(A|E_2)$ = P(gold coin from bag II) = 0

$P(A|E_3)$ = P(gold coin from bag III) $=\frac{1}{2}$

3 Marks

$\therefore P(E_1|A)$ is obtained by Baye's theorem.

$$P(E_1|A)=\frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1)+P(E_2)P(A|E_2)+P(E_3)P(A|E_3)}=\frac{\frac{1}{3}\times 1}{\left(\frac{1}{3}\times 1\right)+\left(\frac{1}{3}\times 0\right)+\left(\frac{1}{3}\times\frac{1}{2}\right)}=\frac{\frac{1}{3}}{\frac{1}{3}+\frac{1}{6}}=\frac{\frac{1}{3}}{\frac{3}{6}}=\frac{2}{3}$$

3 Marks

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22. **In a factory which manufactures bolts, machines A, B and C manufacture respectively 25%, 35% and 40% of the bolts. Of their outputs, 5, 4 and 2 percent are respectively defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it is manufactured by the machine B?**

Sol: Let E_1, E_2 , E_3 be the events of making bolt from machines A,B, C respectively.

Then $P(E_1) = 25\% = 0.25$, P $(E_2) = 35\% = 0.35$ and $P(E_3) = 40\% = 0.40$ **2 Marks**

A	B	C
25%	35%	40%
5%	4%	2%

Let A be the event that 'the bolt is defective' then

$P(A|E_1)$ = P(defective bolt manufactured by machine A)= 5% = 0.05

$P(A|E_2)$ = P(defective bolt manufactured by machine B)=4%= 0.04, **2 Marks**

$P(A|E_3)$ = P(defective bolt manufactured by machine C) =2% = 0.02.

$\therefore P(E_2|A)$ is obtained by Baye's theorem.

$$P(E_2|A)=\frac{P(E_2)P(A|E_2)}{P(E_1)P(A|E_1)+P(E_2)P(A|E_2)+P(E_3)P(A|E_3)}$$

$$=\frac{0.35\times 0.04}{0.25\times 0.05+0.35\times 0.04+0.40\times 0.02}=\frac{0.0140}{0.0345}=\frac{140}{345}=\frac{5\times 28}{5\times 69}=\frac{28}{69}$$ **4 Marks**

Mahesh

23. Coloured balls are distributed in four boxes as shown in the following table:

Box	Colour			
	Black	White	Red	Blue
I	3	4	5	6
II	2	2	2	2
III	1	2	3	1
IV	4	3	1	5

A box is selected at random and then a ball is randomly drawn from the selected box. The colour of the ball is black, what is the probability that ball drawn is from the box III?

Sol: Let E_1, E_2, E_3, E_4 be the events of selecting boxes I, II, III, IV respectively.

$\therefore P(E_1)=P(E_2)=P(E_3)=P(E_4)=\frac{1}{4}$ **2 Marks**

Let A be the event that 'the ball drawn is black'

Box I have 18 balls, Box II have 8 balls, Box III have 7 balls, Box IV have 13 balls

$P(A|E_1)=\frac{3}{18}, P(A|E_2)=\frac{2}{8}, P(A|E_3)=\frac{1}{7}$ and $P(A|E_4)=\frac{4}{13}$ **2 Marks**

$\therefore P(E_3|A)$ is obtained by Baye's theorem.

$$P(E_3|A)=\frac{P(E_3)P(A|E_3)}{P(E_1)P(A|E_1)+P(E_2)P(A|E_2)+P(E_3)P(A|E_3)+P(E_4)P(A|E_4)}$$

$$=\frac{\frac{1}{4}\times\frac{1}{7}}{\left(\frac{1}{4}\times\frac{3}{18}\right)+\left(\frac{1}{4}\times\frac{1}{4}\right)+\left(\frac{1}{4}\times\frac{1}{7}\right)+\left(\frac{1}{4}\times\frac{4}{13}\right)}=\frac{\frac{1}{4}\left(\frac{1}{7}\right)}{\frac{1}{4}\left(\frac{1}{6}+\frac{1}{4}+\frac{1}{7}+\frac{4}{13}\right)}=\frac{\frac{1}{7}}{\frac{947}{1092}}=\frac{156}{947}$$ **4 Marks**

PQ A man is known to speak truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six. **[Ans: 3/ 8]**

PQ In answering a question on a multiple choice test, a student either knows the answer or guesses. Let $\frac{3}{4}$ be the probability that he knows the answer and $\frac{1}{4}$ be the probability that he guesses. Assuming that a student who guesses at the answer will be correct with probability $\frac{1}{4}$. What is the probability that the student knows the answer given that he answered it correctly? **[Ans: 12/ 13]**

[For Answers ref. QR Code on P.No.52]

Mahesh

24. Suppose we have four boxes A,B,C, D containing coloured marbles as given below: **[BMP]**

Box	Marble colour		
	Red	White	Black
A	1	6	3
B	6	2	2
C	8	1	1
D	0	6	4

One of the boxes has been selected at random and a single marble is drawn from it.

If the marble is red, what is the probability that it was drawn from box A?, box B?, box C?

Sol: Let E_1, E_2 , E_3 be the events of selecting the boxes A, B, C respectively.
Let A be the event of drawing the red marble.Number of red marbles = 15.
Total number of marbles = 15 R + 15 W + 10 B = 40

$$\therefore P(A)=\frac{15}{40}=\frac{3}{8}$$

2 Marks

Probability of drawing the red marble from box A is given by $P(E_1\cap A)=\frac{1}{40}$

Probability that the red marble is from box A is $P(E_1|A)$.

$$\therefore P(E_1\,|\,A)=\frac{P(E_1\cap A)}{P(A)}=\frac{\frac{1}{40}}{\frac{3}{8}}=\frac{1}{\cancel{40}}\times\frac{\cancel{8}}{3}=\frac{1}{15}$$

2 Marks

Probability of drawing the red marble from box B is given by $P(E_2\cap A)=\frac{6}{40}$
Probability that the red marble is from box B is $P(E_2|A)$.

$$\therefore P(E_2\,|\,A)=\frac{P(E_2\cap A)}{P(A)}=\frac{\frac{6}{40}}{\frac{3}{8}}=\frac{\cancel{6}}{\cancel{40}}\times\frac{\cancel{8}}{\cancel{3}}=\frac{2}{5}$$

2 Marks

Probability of drawing the red marble from box C is given by $P(E_3\cap A)=\frac{8}{40}$

Probability that the red marble is from box C is $P(E_3|A)$.

$$\therefore P(E_3\,|\,A)=\frac{P(E_3\cap A)}{P(A)}=\frac{\frac{8}{40}}{\frac{3}{8}}=\frac{\cancel{8}}{\cancel{40}}\times\frac{8}{3}=\frac{8}{15}$$

2 Marks

Mahesh

5 APPLICATION OF DERIVATIVES

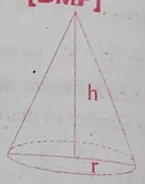
8 Marks

PROBLEMS ON RATE MEASURE

25. **Sand is pouring from a pipe at the rate of 12 cm^3/s. The falling sand forms a cone on the ground in such a way that the height of the cone is always one-sixth of the radius of the base. How fast is the height of the sand cone increasing when the height is 4 cm?**

[BMP]

Sol: Consider a cone of radius r, height h and volume V



Given that $h=\frac{1}{6}r \Rightarrow r=6h$, $\frac{dV}{dt}=12cm^2/s$ and $h=4cm$ — 2 Mark

Volume of cone is $V=\frac{\pi}{3}r^2h \Rightarrow V=\frac{\pi}{3}(6h)^2h=12\pi h^3$ — 2 Mark

$\Rightarrow \frac{dV}{dt}=12\pi\left(3h^2\frac{dh}{dt}\right)=36\pi h^2\frac{dh}{dt}$ — 2 Mark

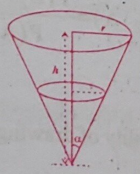
$\Rightarrow 12=36\pi(4)^2\frac{dh}{dt} \Rightarrow \frac{dh}{dt}=\frac{12}{36\pi(16)}=\frac{1}{48\pi}cm/s \quad \left[\because \frac{dV}{dt}=12, h=4\right]$ — 2 Mark

---

26. **A water tank has the shape of an inverted right circular cone with its axis vertical and vertex lowermost. Its semi-vertical angle is $\tan^{-1}(0.5)$. Water is poured into it at a constant rate of 5 cubic metre per hour. Find the rate at which the level of the water is rising at the instant when the depth of water in the tank is 4 m.**

Sol: Consider a cone of radius r, height h, semi-vertical angle α and volume V

Given $\frac{dV}{dt}=5m/h$ and $h=4$ m then



2 Mark

Now $\tan\alpha=\frac{r}{h} \Rightarrow \alpha=\tan^{-1}\left(\frac{r}{h}\right)$

Given $\alpha=\tan^{-1}(0.5) \Rightarrow \tan\alpha=0.5=\frac{1}{2} \Rightarrow \frac{r}{h}=\frac{1}{2} \Rightarrow r=\frac{h}{2}$ — 2 Mark

Volume of cone is $V=\frac{\pi}{3}r^2h=\frac{\pi}{3}\left(\frac{h}{2}\right)^2h=\frac{\pi h^3}{12} \Rightarrow \frac{dV}{dt}=\frac{\pi}{12}\left(3h^2\frac{dh}{dt}\right)=\frac{\pi}{4}h^2\frac{dh}{dt}$ — 2 Mark

$\Rightarrow 5=\frac{\pi}{4}(4)^2\frac{dh}{dt} \Rightarrow \frac{dh}{dt}=\frac{5}{4}\left(\frac{7}{22}\right)=\frac{35}{88} \quad \left(\because \frac{dV}{dt}=5, h=4, \pi=\frac{22}{7}\right)$ — 2 Mark

Thus, the rate of change of water level is $\frac{35}{88}$ m/h

Naresh

PROBLEMS ON MAXIMA & MINIMA

27. Find two positive numbers whose sum is 15 and the sum of whose squares is minimum.

Sol: Let the two positive numbers be x,y

1) Given that $x + y = 15 \Rightarrow y = 15 - x$.....(1)

2) Let $f(x) = x^2 + y^2 = x^2 + (15 - x)^2$

$\therefore f(x) = x^2 + (15 - x)^2$.....(2) [$\because$ From (1)]

3) $\Rightarrow f'(x) = 2x + 2(15 - x)(-1)$

$= 2x - 30 + 2x = 4x - 30 = 2(2x - 15)$ ....(3)

4) At max. or min. we have $f'(x) = 0$

$\Rightarrow 2x - 15 = 0 \Rightarrow x = 15/2$

5) Now, Diff. (3) w.r.t x,we get $f''(x) = 4$ ...(4)

6) At $x = 15/2$, from (4), $f''(15/2) = 4 > 0$

7) $\therefore f(x)$ is minimum when $x = 15/2$ and $y = 15 - \frac{15}{2} = \frac{15}{2}$

8) Hence required numbers are $x = \frac{15}{2}, y = \frac{15}{2}$

SOLUTION STEPS

Objective function: $x^2 + y^2$

1. Write the relation between x & y.
2. Write f(x)
3. Find f'(x)
4. Take f'(x)=0 to get x= a, b
5. Find f''(x)
6. Find the sign of f''(a), f''(b)

7.1 If f''(a)<0 then f(a) is max.

7.2 If f''(a)>0 then f(a) is min.

PQ **Find two positive numbers whose sum is 16 and the sum of whose cubes is minimum.**
[For Answer ref. QR Code on P.No.52] [Ans: 8,8]

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28. Find two positive numbers x and y such that x+y=60 and xy^3 is maximum.

Sol: 1) Given $x + y = 60 \Rightarrow x = 60 - y$.....(1)

2) Let $f(y) = xy^3$

From (1), $f(y) = (60 - y)y^3 \therefore f(y) = 60y^3 - y^4$ .....(2)

3) On Diff. (2), w.r.to 'y', $f'(y) = 60(3y^2) - 4y^3$

$\Rightarrow f'(y) = 180y^2 - 4y^3$.....(3)

4) $f'(y) = 0 \Rightarrow 180y^2 - 4y^3 = 0 \Rightarrow 4y^2(45 - y) = 0$

$\Rightarrow y = 0$ (or) $y = 45$

Also from (1), $x = 60 - y = 60 - 45 = 15$

5) From (3), $f''(y) = 360y - 12y^2 = 12y(30 - y)$ .....(4)

6) At y=45, from (4) we get

$f''(45) = 12(45)(30 - 45) = 12(45)(-15) < 0$

7) $\therefore f(y)$ has a maximum value at y=45 and x=15

8) Hence, the required integers are 15, 45.

☺ To have easy calculations we took f(y) instead of f(x)

SOLUTION STEPS

Objective function: xy^3

1. Write the relation between x & y.
2. Write f(y)
3. Find f'(y)
4. Take f'(y)=0 to get y= a, b
5. Find f''(y)
6. Find the sign of f''(a), f''(b)

7.1 If f''(a)<0 then f(a) is max.

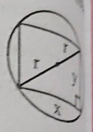
7.2 If f''(a)>0 then f(a) is min.

Mahesh

29. **Show that of all the rectangles inscribed in a given fixed circle, the square has maximum area.**

Sol: Let the sides of the rectangle be x,y and r be the radius of the circle(r is a constant)

1) From the diagram, $x^2+y^2=(2r)^2 \Rightarrow 4r^2=x^2+y^2$ ...(1)



Area of the rectangle is $A=xy=x\sqrt{4r^2-x^2}$

2) $A^2(x)=x^2(4r^2-x^2)=4r^2x^2-x^4=f(x)$ (say)

3) $f'(x)=4r^2(2x)-4x^3=8r^2x-4x^3$

4) $\Rightarrow f'(x)=0 \Rightarrow 8r^2x-4x^3=0 \Rightarrow 4x(2r^2-x^2)=0$ 4 Marks

$x^2=2r^2 \Rightarrow x=\sqrt{2}r$

5) $f''(x)=8r^2-12x^2$

6) $f''(\sqrt{2}r)=8r^2-12(\sqrt{2}r)^2=8r^2-12(2r^2)=8r^2-24r^2=-16r^2<0$

7) $\therefore A(x)$ is maximum when $x=y=\sqrt{2}r$

8) The dimensions of the rectangle are $x=y=\sqrt{2}r$ which form a square.
$\therefore$ the maximum rectangle that can be inscribed in a circle is a square 4 Marks

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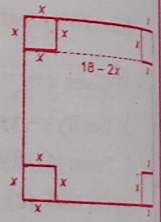
30. **A square piece of tin of side 18 cm is to be made into a box without top, by cutting a square from each corner and folding up the flaps to form the box. What should the side of the square to be cut off so that the volume of the box is the maximum possible.**

Sol: 1) Let each side of the square to be cut off be x cm.



For the box, length $l=18-2x$, $b=18-2x$, $h=x$

2) $\therefore$ Volume of the box is $V(x)=lbh=(18-2x)^2x$

3) $\Rightarrow V'(x)=(18-2x)^2(1)+x.2(18-2x)(-2)$

$=(18-2x)^2-4x(18-2x)=(18-2x)[18-2x-4x]$

$=(18-2x)(18-6x)=2(9-x)6(3-x)=12(9-x)(3-x)$

$=12(x-9)(x-3)=12(x^2-12x+27)$ 4 Marks

4) Now, $V'(x)=0 \Rightarrow x=9$ or 3

If $x=9$ then the length and the breadth 18–2x will become 0. Hence $x\neq 9$

5) $V''(x)=12(2x-12)=24(x-6)$.

6) When x=3 then V"(3)= 24(3–6)= –72<0

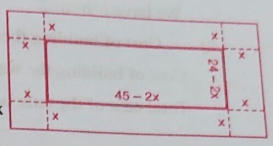
7) By second derivative test, x=3 is the point of local maxima of V.

8) Hence, to get maximum volume, we have to remove a square of side 3 cm from each corner 4 Marks

Naresh

31. **A rectangular sheet of tin 45 cm by 24 cm is to be made into a box without top, by cutting off square from each corner and folding up the flaps. What should be the side of the square to be cut off so that the volume of the box is maximum ?**

Sol: 1) Let each side of the square to be cut off be xcm.



For the box, length $l = 45-2x$, $b = 24-2x$, $h = x$

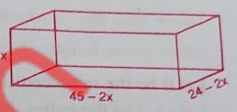
$\therefore$ Volume of the box is

2) $V(x) = (45-2x)(24-2x)x = (1080-90x-48x+4x^2)x$

$= 4x^3 - 138x^2 + 1080x$

3) $\Rightarrow V'(x) = 12x^2 - 276x + 1080 = 12(x^2 - 23x + 90)$

$= 12(x-18)(x-5)$



4) Now, $V'(x) = 0 \Rightarrow x = 18$ or 5

If $x=18$ then breadth $b = 24 - 2(18) = -12$ which is not possible.

5) $\Rightarrow V''(x) = 24x - 276 = 12(2x-23)$

6) $\therefore$ When $x=5$ then $V''(5) = 12[2(5)-23] = 12(10-23) = 12(-13) = -156 < 0$

7) By second derivative test, $x=5$ is the point of local maxima.

8) Hence, to get maximum volume, we have to remove a square of side 5 cm from each corner.

PQ **An open topped box is to be constructed by removing equal squares from each corner of a 3 metre by 8 metre rectangular sheet of aluminium and folding up the sides. Find the volume of the largest such box. [Ans: 200/27 m^3]**

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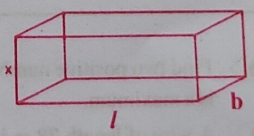
32. **A tank with rectangular base and rectangular sides, open at the top is to be constructed so that its depth is 2 m and volume is 8 m^3. If building of tank costs Rs 70 per sq metres for the base and Rs 45 per square metre for sides. What is the cost of least expensive tank?**

Sol: 1) Let l,b, h represent the length, breadth, and height of the tank respectively.

Then we have height $h=2m$ and volume of the tank $V=8m^3$

Volume of the tank $V = lbh \Rightarrow 8 = l \times b \times 2 \Rightarrow lb = 4 \Rightarrow b = \frac{4}{l}$

Now, area of the base is $lb=4$



2) Area of the 4 walls $A(l) = 2h(l+b) = 4\left(l + \frac{4}{l}\right)$

3) $A'(l) = 4\left(1 - \frac{4}{l^2}\right)$

4) Now, $A'(l) = 0 \Rightarrow \left(1 - \frac{4}{l^2}\right) = 0 \Rightarrow l^2 = 4 \Rightarrow l = \pm 2 \Rightarrow l = 2$

Hence, $b = \frac{4}{l} = \frac{4}{2} = 2$

5) $A''(l) = -16\left(\frac{-2}{l^3}\right) = \frac{32}{l^3}$ $\left[\because A'(l) = 4 - \frac{16}{l^2}\right]$

4 Marks

Mahesh

6) When, $l=2$, $A''(l)=\frac{32}{8}=4>0$

7) Thus, by second derivative test, the area is the minimum when $l=2$

We have $l=b=h=2$

8) ∴ Cost of building the base at 70 Rs is $(lb)70=(2)(2)70=$ Rs 280

Cost of building the walls at 45 Rs is $2h(l+b)(45)=(2\times2)(2+2)(45)=$ Rs 720

Total cost of the tank is 280+720= Rs 1000

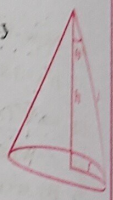
4 Ma

33. Show that the semi-vertical angle of the cone of the maximum volume and of given slant height is $\tan^{-1}\sqrt{2}$

Sol: Let θ be the semi-vertical angle of the cone.

Let r,h, l be the radius, height, and the slant height of the cone respectively

The slant height l of the cone is given as 'constant'.



1) From diagram, $\tan\theta=\frac{r}{h}$; $l^2=r^2+h^2$.....(1)

2) Volume of the cone is $V=\frac{\pi}{3}r^2h=\frac{\pi}{3}h(l^2-h^2)=\frac{\pi}{3}(l^2h-h^3)$

$\Rightarrow V(h)=\frac{\pi}{3}(l^2h-h^3)$

4 Mar

3) $V'(h)=\frac{\pi}{3}(l^2-3h^2)$

4) For Max or Min. $V'(h)=0 \Rightarrow l^2-3h^2=0\Rightarrow l^2=3h^2$

5) From (1) $3h^2=r^2+h^2\Rightarrow 2h^2=r^2\Rightarrow\frac{r^2}{h^2}=2\Rightarrow\frac{r}{h}=\sqrt{2}\Rightarrow\tan\theta=\sqrt{2}\Rightarrow\theta=\tan^{-1}\sqrt{2}$

6) $V''(h)=\frac{\pi}{3}(-6h)=-2\pi h<0$

7) Hence the semi-vertical angle of the cone of the maximum volume is $\tan^{-1}\sqrt{2}$

4 Mar

Mahesh

ADDITIONAL PQs

PQ Find two positive numbers x and y such that their sum is 35 and the product x^2y^5 is a maximum.

PQ A wire of length 28 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into a circle. What should be the length of the two pieces so that the combined area of the square and the circle is minimum?

PQ Show that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius R is $\frac{2R}{\sqrt{3}}$. Also find the maximum volume.

[For Answer ref. QR Code on P.No.52

8 + 8 Marks

6 INTEGRATION

6 DI + 4 PF + 6 QF

PROBLEMS ON DEFINITE INTEGRALS

34. Evaluate $\int_0^{\pi} \frac{x}{1+\sin x}\,dx$ **[BMP]**

Sol: Let $I = \int_0^{\pi} \frac{x}{1+\sin x}dx = \int_0^{\pi} \frac{(\pi - x)dx}{1+\sin(\pi - x)}$ $\left[\because \int_0^a f(x)dx = \int_0^a f(a-x)dx\right]$

$$\therefore I = \int_0^{\pi} \frac{\pi\,dx}{1+\sin x} - \int_0^{\pi} \frac{x\,dx}{1+\sin x} = \int_0^{\pi} \frac{\pi\,dx}{1+\sin x} - I \Rightarrow I + I = 2I = \pi\int_0^{\pi} \frac{dx}{1+\sin x}$$ **2 Marks**

$$\Rightarrow I = \frac{\pi}{2}\int_0^{\pi} \frac{dx}{1+\sin x} = \frac{\pi}{2}\int_0^{\pi} \frac{(1-\sin x)}{1-\sin^2 x}dx = \frac{\pi}{2}\int_0^{\pi}\left(\frac{1-\sin x}{\cos^2 x}\right)dx$$ **2 Marks**

$$= \frac{\pi}{2}\left(\int_0^{\pi} \frac{1}{\cos^2 x}dx - \int_0^{\pi} \frac{\sin x}{\cos x}\cdot\frac{1}{\cos x}dx\right) = \frac{\pi}{2}\int_0^{\pi} \sec^2 x\,dx - \int_0^{\pi} \sec x.\tan x\,dx$$

$$= \frac{\pi}{2}\left[[\tan x]_0^{\pi} - [\sec x]_0^{\pi}\right] = \frac{\pi}{2}[(0-0)-(-1-1)] = \frac{\pi}{2}(2) = \pi$$ **4 Marks**

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35. Evaluate $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\,dx$

Sol: We know $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ $\therefore I = \int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}dx = \int_0^{\pi} \frac{(\pi - x)\sin(\pi - x)}{1+\cos^2(\pi - x)}dx$

$$= \int_0^{\pi} \frac{(\pi - x)\sin x}{1+\cos^2 x}dx = \int_0^{\pi} \frac{\pi\sin x}{1+\cos^2 x}dx - \int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}dx = \int_0^{\pi} \frac{\pi\sin x}{1+\cos^2 x}dx - I$$

$$\Rightarrow I + I = 2I = \int_0^{\pi} \frac{\pi\sin x}{1+\cos^2 x}dx \Rightarrow 2I = \pi\int_0^{\pi} \frac{\sin x}{1+\cos^2 x}dx \Rightarrow I = \frac{\pi}{2}\int_0^{\pi} \frac{\sin x}{1+\cos^2 x}dx$$ **4 Marks**

Put $\cos x = t \Rightarrow \sin x\,dx = -dt$

Also, $x = 0 \Rightarrow t = \cos 0 = 1$ and $x = \pi \Rightarrow t = \cos\pi = -1$

$$\therefore I = \frac{\pi}{2}\int_1^{-1} \frac{-dt}{1+t^2} = \frac{\pi}{2}\int_{-1}^{1} \frac{dt}{1+t^2} = \frac{\pi}{\cancel{2}}\cancel{2}\int_0^{1} \frac{dt}{1+t^2} = \pi[\mathrm{Tan}^{-1}t]_0^1 \quad \left[\because \frac{1}{1+t^2} \text{ is an even function}\right]$$

$$= \pi[\mathrm{Tan}^{-1}(1) - \mathrm{Tan}^{-1}(0)] = \pi\left(\frac{\pi}{4} + 0\right) = \frac{\pi^2}{4}$$ **4 Marks**

Mahesh

36. **Evaluate** $\int_0^{\pi/4} \log(1+\tan x)\,dx$

Sol: Let $I = \int_0^{\pi/4} \log(1+\tan x)\,dx$ $\left[\because \int_0^a f(x)dx = \int_0^a f(a-x)dx\right]$

$$\therefore I = \int_0^{\pi/4} \log\left[1+\tan\left(\frac{\pi}{4}-x\right)\right]dx = \int_0^{\pi/4} \log\left[1+\frac{\tan\frac{\pi}{4}-\tan x}{1+\tan\frac{\pi}{4}\tan x}\right]dx$$

4 Mark

$$= \int_0^{\pi/4} \log\left(1+\frac{1-\tan x}{1+\tan x}\right)dx = \int_0^{\pi/4} \log\left(\frac{(1+\cancel{\tan x})+(1-\cancel{\tan x})}{1+\tan x}\right)dx = \int_0^{\pi/4} \log\left(\frac{2}{1+\tan x}\right)dx$$

$$= \int_0^{\pi/4} [\log 2 - \log(1+\tan x)]dx = \log 2 \int_0^{\pi/4} 1dx - \int_0^{\pi/4} \log(1+\tan x)\,dx = \log 2\,[x]_0^{\pi/4} - I$$

$$\Rightarrow I+I = 2I = \log 2\left(\frac{\pi}{4}\right) \Rightarrow I = \frac{\pi}{8}\log 2$$

4 Mark

---

37. **Evaluate** $\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx$

Sol: Put $x = \tan\theta \Rightarrow dx = \sec^2\theta d\theta$. Also, $x = 0 \Rightarrow \theta = 0; x = 1 \Rightarrow \theta = \frac{\pi}{4}$

Now, $1+x^2 = 1+\tan^2\theta = \sec^2\theta$

$$\therefore \int_0^1 \frac{\log(1+x)}{1+x^2}dx = \int_0^{\pi/4} \frac{\log(1+\tan\theta)}{\cancel{\sec^2\theta}}\cancel{\sec^2\theta}\,d\theta = \int_0^{\pi/4} \log(1+\tan\theta)d\theta$$

[Same as previou

Let $I = \int_0^{\pi/4} \log[1+\tan\theta]d\theta = \int_0^{\pi/4} \log\left[1+\tan\left(\frac{\pi}{4}-\theta\right)\right]d\theta = \int_0^{\pi/4} \log\left[1+\frac{\tan\frac{\pi}{4}-\tan\theta}{1+\tan\frac{\pi}{4}\tan\theta}\right]d\theta$

4 Mar

$$= \int_0^{\pi/4} \log\left[1+\frac{1-\tan\theta}{1+\tan\theta}\right]d\theta = \int_0^{\pi/4} \log\left[\frac{(1+\cancel{\tan\theta})+(1-\cancel{\tan\theta})}{1+\tan\theta}\right]d\theta = \int_0^{\pi/4} \log\left(\frac{2}{1+\tan\theta}\right)d\theta$$

$$= \int_0^{\pi/4} [\log 2 - \log(1+\tan\theta)]\,d\theta = \int_0^{\pi/4} \log 2d\theta - \int_0^{\pi/4} \log(1+\tan\theta)d\theta = \int_0^{\pi/4} \log 2d\theta - I$$

$$\Rightarrow I+I = 2I = \int_0^{\pi/4} \log 2d\theta = \log 2[\theta]_0^{\pi/4} = \frac{\pi}{4}\log 2 \Rightarrow I = \frac{\pi}{8}\log 2$$

4 Mar

Mahesh

38. **Evaluate** $\int_0^{\pi/4} \frac{\sin x+\cos x}{9+16\sin 2x}dx$

Sol: Here, we take the substitution sinx−cosx=t. Then (cosx+sinx)dx=dt.

Now $x=0 \Rightarrow t=\sin 0-\cos 0 = 0-1=-1$ and $x=\pi/4 \Rightarrow t=\sin\frac{\pi}{4}-\cos\frac{\pi}{4}=\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{2}}=0$

Also,$(\sin x-\cos x)^2=t^2 \Rightarrow \sin^2x+\cos^2x-2\sin x\cos x=t^2 \Rightarrow 1-\sin 2x=t^2 \Rightarrow \sin 2x=1-t^2$

$\therefore 9+16\sin 2x=9+16(1-t^2)=9+16-16t^2=25-16t^2$ **4 Marks**

$$\therefore I=\int_0^{\pi/4}\frac{\sin x+\cos x}{9+16\sin 2x}dx=\int_{-1}^{0}\frac{dt}{25-16t^2}=\int_{-1}^{0}\frac{dt}{5^2-(4t)^2}=\frac{1}{4}\cdot\frac{1}{2(5)}\cdot\log\left[\frac{5+4t}{5-4t}\right]_{-1}^{0}$$

$$=\frac{1}{40}\left[\log\left[\frac{5+0}{5-0}\right]-\log\left[\frac{5-4}{5+4}\right]\right]=\frac{1}{40}\left[\log 1-\log\frac{1}{9}\right]$$

$$=\frac{1}{40}[0-\log 9^{-1}]=\frac{1}{40}[\log 9]=\frac{\log 3^2}{40}=\frac{2\log 3}{40}=\frac{\log 3}{20}$$ **4 Marks**

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39. **Evaluate** $\int_0^{\frac{\pi}{2}}\log\sin x dx$

Sol: Let $I=\int_0^{\frac{\pi}{2}}\log\sin x dx$.....(1) $\left[\because \int_0^a f(x)dx=\int_0^a f(a-x)dx\right]$

$\Rightarrow I=\int_0^{\frac{\pi}{2}}\log\sin\left(\frac{\pi}{2}-x\right)dx=\int_0^{\frac{\pi}{2}}\log\cos x dx$.....(2) Adding (1) and (2) we get

$$2I=\int_0^{\frac{\pi}{2}}(\log\sin x+\log\cos x)dx=\int_0^{\frac{\pi}{2}}(\log\sin x\cos x)dx$$

$$=\int_0^{\frac{\pi}{2}}\left(\log\frac{2\sin x\cos x}{2}\right)dx=\int_0^{\frac{\pi}{2}}(\log 2\sin x\cos x-\log 2)dx$$

$$=\int_0^{\frac{\pi}{2}}\log\sin 2x dx-\int_0^{\frac{\pi}{2}}\log 2dx=\int_0^{\frac{\pi}{2}}\log\sin 2x-\frac{\pi}{2}\log 2 \quad \left[\because \int_0^{\frac{\pi}{2}}\log 2dx=\log 2[x]_0^{\pi/2}=\log 2\left(\frac{\pi}{2}\right)\right]$$

4 Marks

Put 2x = t in the first integral. Then $2dx=dt \Rightarrow dx=\frac{dt}{2}$

When x = 0, t = 0 and when $x=\frac{\pi}{2}$, t=π

$$\therefore 2I=\frac{1}{2}\int_0^{\pi}\log\sin t dt-\frac{\pi}{2}\log 2=\frac{2}{2}\int_0^{\frac{\pi}{2}}\log\sin t dt-\frac{\pi}{2}\log 2 \qquad [\because \sin(\pi-t)=\sin t]$$

$$=\int_0^{\frac{\pi}{2}}\log\sin x dx-\frac{\pi}{2}\log 2 \qquad \text{(by changing variable t to x)}$$

$\therefore 2I=I-\frac{\pi}{2}\log 2 \Rightarrow 2I-I=-\frac{\pi}{2}\log 2 \Rightarrow I=-\frac{\pi}{2}\log 2=\frac{\pi}{2}\log 2^{-1}=\frac{\pi}{2}\log\frac{1}{2}$ **4 Marks**

PQ **Evaluate** $\int_0^{\pi}\log(1+\cos x)dx$ $\left[\text{Ans}:\pi\log\frac{1}{2}\right]$

Manesh

PROBLEMS ON INDEFINITE INTEGRALS

Partial Fraction Questions (Q 40 to 43)

40. **Find the integral of** $\frac{x^2+x+1}{(x+1)^2(x+2)}$

Sol: Let $\frac{x^2+x+1}{(x+1)^2(x+2)}=\frac{A}{(x+1)}+\frac{B}{(x+1)^2}+\frac{C}{(x+2)}=\frac{A(x+1)(x+2)+B(x+2)+C(x+1)^2}{(x+1)^2(x+2)}$ **1 Mark**

$\Rightarrow A(x+1)(x+2)+B(x+2)+C(x+1)^2=x^2+x+1$.....(1)

$=A(x^2+3x+2)+B(x+2)+C(x^2+2x+1)$

$\Rightarrow x^2+x+1=(A+C)x^2+(3A+B+2C)x+(2A+2B+C)$

Put $x=-1$ in (1) $\Rightarrow A(0)+B(-1+2)+C(0)=1-1+1\Rightarrow B=1$

Put $x=-2$ in (1) $\Rightarrow A(0)+B(0)+C(-2+1)^2=(-2)^2-2+1\Rightarrow C=4-2+1=3$

Equating the coefficients of x^2, we get $A+C=1\Rightarrow A=1-C=1-3=-2$ **3 Marks**

$\therefore A=-2, B=1, C=3$

From (1), we get $\frac{x^2+x+1}{(x+1)^2(x+2)}=\frac{-2}{(x+1)}+\frac{3}{(x+2)}+\frac{1}{(x+1)^2}$

$$\int\frac{x^2+x+1}{(x+1)^2(x+2)}dx=-2\int\frac{1}{x+1}dx+3\int\frac{1}{(x+2)}dx+\int\frac{1}{(x+1)^2}dx$$

$$=-2\log|x+1|+3\log|x+2|-\frac{1}{(x+1)}+C$$ **4 Marks**

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41. **Find** $\int\frac{x^2+x+1dx}{(x+2)(x^2+1)}$

Sol: Let $\frac{x^2+x+1}{(x^2+1)(x+2)}=\frac{A}{x+2}+\frac{Bx+C}{(x^2+1)}=\frac{A(x^2+1)+(Bx+C)(x+2)}{(x+2)(x^2+1)}$ **1 Mark**

$\Rightarrow A(x^2+1)+(Bx+C)(x+2)=x^2+x+1$.....(1)

Put $x=-2$ in (1) $\Rightarrow A(4+1)+0=4-2+1\Rightarrow 5A=3\Rightarrow A=\frac{3}{5}$

Equating the coefficients of x^2, we get $A+B=1\Rightarrow B=1-A=1-\frac{3}{5}=\frac{2}{5}$

Equating the coefficients of x, we get $2B+C=1\Rightarrow C=1-2B=1-2\left(\frac{2}{5}\right)=1-\frac{4}{5}=\frac{1}{5}$

Now $\frac{x^2+x+1}{(x^2+1)(x+2)}=\frac{3}{5(x+2)}+\frac{\frac{2}{5}x+\frac{1}{5}}{x^2+1}=\frac{3}{5(x+2)}+\frac{1}{5}\left(\frac{2x+1}{x^2+1}\right)$ **3 Marks**

$$\therefore\int\frac{x^2+x+1}{(x^2+1)(x+2)}dx=\frac{3}{5}\int\frac{dx}{x+2}+\frac{1}{5}\int\frac{2x}{x^2+1}dx+\frac{1}{5}\int\frac{1}{x^2+1}dx$$

$$=\frac{3}{5}\log|x+2|+\frac{1}{5}\log|x^2+1|+\frac{1}{5}\tan^{-1}x+C$$ **4 Marks**

Manesh

42. Find $\int \frac{x^4 dx}{(x-1)(x^2+1)}$

Sol: In integrand, the degree of Nr > Degree of Dr.

On dividing x^4 by $(x-1)(x^2+1) = x^3-x^2+x-1$ we get the quotient $x+1$

$$\frac{x^4}{(x-1)(x^2+1)} = (x+1) + \frac{1}{x^3-x^2+x-1} = (x+1) + \frac{1}{(x-1)(x^2+1)}$$ **2 Marks**

Now $\frac{1}{(x-1)(x^2+1)} = \frac{A}{(x-1)} + \frac{Bx+C}{(x^2+1)} = \frac{A(x^2+1)+(Bx+C)(x-1)}{(x-1)(x^2+1)}$

$\Rightarrow A(x^2+1)+(Bx+C)(x-1) = (A+B)x^2+(C-B)x+(A-C)=1$.....(1)

Put x=1 in (1) $\Rightarrow$ A(1+1)=1$\Rightarrow$A=1/2

Equating the coefficients of x^2, we get A+B=0$\Rightarrow$B = –A = – 1/2

Equating the coefficients of x, we get C–B=0$\Rightarrow$C=B – 1/2 $\Rightarrow$ C=$\frac{-1}{2}$ **2 Marks**

$$\therefore \frac{1}{(x-1)(x^2+1)} = \frac{1}{2(x-1)} - \frac{1}{2}\frac{x}{(x^2+1)} - \frac{1}{2(x^2+1)}$$

Now, $$\frac{x^4}{(x-1)(x^2+1)} = (x+1) + \frac{1}{2(x-1)} - \frac{1}{2}\frac{x}{(x^2+1)} - \frac{1}{2(x^2+1)}$$

$$\therefore \int \frac{x^4}{(x-1)(x^2+1)}dx = \frac{x^2}{2} + x + \frac{1}{2}\log|x-1| - \frac{1}{4}\log(x^2+1) - \frac{1}{2}\tan^{-1}x + C$$ **4 Marks**

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43. Find $\int \frac{x^2}{(x^2+1)(x^2+4)}dx$

Sol: Put $x^2 = y$ in the given integrand then $\frac{x^2}{(x^2+1)(x^2+4)} = \frac{y}{(y+1)(y+4)}$ **1 Mark**

Let $\frac{y}{(y+1)(y+4)} = \frac{A}{y+1} + \frac{B}{y+4} = \frac{A(y+4)+B(y+1)}{y+1}$

$\Rightarrow A(y+4) + B(y+1) = y$ ....(1)

Put y= –1 in (1) $\Rightarrow$ 3A+0= –1 $\Rightarrow$ 3A = –1 $\Rightarrow$ A=$\frac{-1}{3}$ **2 Marks**

Put y= –4 in (1) $\Rightarrow$ 0–3B = –4 $\Rightarrow$ 3B = 4 $\Rightarrow$ B=$\frac{4}{3}$ **1 Mark**

Thus, $$\frac{x^2}{(x^2+1)(x^2+4)} = \frac{-1}{3(x^2+1)} + \frac{4}{3(x^2+4)}$$

$$\therefore \int \frac{x^2dx}{(x^2+1)(x^2+4)} = -\frac{1}{3}\int \frac{dx}{x^2+1} + \frac{4}{3}\int \frac{dx}{x^2+2^2} \left[\because \int \frac{1}{x^2+a^2}dx = \tan^{-1}\left(\frac{x}{a}\right)+C\right]$$

$$= -\frac{1}{3}\tan^{-1}x + \frac{4}{3}\left(\frac{1}{2}\tan^{-1}\frac{x}{2}\right) + C = -\frac{1}{3}\tan^{-1}x + \frac{2}{3}\tan^{-1}\frac{x}{2} + C$$ **4 Marks**

Manish

Quadratic Functions in Integrands (Q 44 to 48)

44. Find $\int \frac{x+2}{2x^2+6x+5}dx$

Sol: Let $x+2 = A\frac{d}{dx}(2x^2+6x+5)+B = A(4x+6)+B$ ....(1)

Equating the coefficients of x on both sides, we get $4A = 1 \Rightarrow A = \frac{1}{4}$

Equating the constant terms, we get $6A+B=2 \Rightarrow B = 2-6A = 2-\frac{6}{4} = 2-\frac{3}{2} = \frac{1}{2}$ 2 Marks

Hence from (1), $(x+2) = \frac{1}{4}(4x+6)+\frac{1}{2}$

2 Marks

$$\therefore \int \frac{x+2}{2x^2+6x+5} = \int \frac{\frac{1}{4}(4x+6)+\frac{1}{2}}{2x^2+6x+5}dx = \frac{1}{4}\int \frac{4x+6}{2x^2+6x+5}dx + \frac{1}{2}\int \frac{dx}{2x^2+6x+5}$$

$$= \frac{1}{4}\log|2x^2+6x+5| + \frac{1}{2}\int \frac{dx}{2x^2+6x+5} \quad .....(2) \qquad \left[\because \int \frac{f'(x)}{f(x)}dx = \log|f(x)|\right]$$

Let $I = \int \frac{dx}{2x^2+6x+5} = \frac{1}{2}\int \frac{dx}{x^2+3x+\frac{5}{2}} = \frac{1}{2}\int \frac{dx}{\left(x+\frac{3}{2}\right)^2+\left(\frac{1}{2}\right)^2}$ 2 Marks

$$= \frac{1}{2}\tan^{-1}\left(\frac{1}{1/2}\right)\left(x+\frac{3}{2}\right) = \frac{1}{2}\tan^{-1}2\left(x+\frac{3}{2}\right) = \frac{1}{2}\tan^{-1}(2x+3) \qquad \left[\because \int \frac{1}{x^2+a^2}dx = \frac{1}{a}\left(\tan^{-1}\frac{x}{a}\right)\right]$$

From (2), $\int \frac{x+2}{2x^2+6x+5}dx = \frac{1}{4}\log\left|2x^2+6x+5\right| + \frac{1}{2}\tan^{-1}(2x+3)+C$ 2Marks

☺ HINT BOX ☺

'COMPLETING THE SQUARE' PROCESS OF A QUADRATIC

Reducing process of ax^2+bx+c into $A^2 \pm X^2$ form.

$$ax^2+bx+c = a\left(x^2+\frac{b}{a}x+\frac{c}{a}\right) = a\left(x^2+\frac{b}{a}x+\left(\frac{b}{2a}\right)^2-\left(\frac{b}{2a}\right)^2+\frac{c}{a}\right)$$

Q44: $2x^2+6x+5 = 2\left(x^2+3x+\frac{5}{2}\right) = 2\left(x^2+2\cdot\frac{3}{2}x+\left(\frac{3}{2}\right)^2-\left(\frac{3}{2}\right)^2+\frac{5}{2}\right)$

Manesh

45. **Find the integral of** $\frac{x+3}{x^2-2x-5}$ **[BMP]**

Sol: Let $(x+3)=A\frac{d}{dx}(x^2-2x-5)+B=A(2x-2)+B$ .....(1)

Equating the coefficients of x on both sides, we get $2A=1\Rightarrow A=\frac{1}{2}$

Equating the constant terms, we get $-2A+B=3\Rightarrow B=3+2A=3+2\left(\frac{1}{2}\right)=3+1=4$

Hence from (1), $(x+3)=\frac{1}{2}(2x-2)+4$ **2 Marks**

$$\therefore\int\frac{x+3}{x^2-2x-5}dx=\int\frac{\frac{1}{2}(2x-2)+4}{x^2-2x-5}dx=\frac{1}{2}\int\frac{2x-2}{x^2-2x-5}dx+4\int\frac{1}{x^2-2x-5}dx$$ **2 Marks**

$$=\frac{1}{2}\log\left|x^2-2x-5\right|+4\int\frac{dx}{x^2-2x-5}\text{ .....(2)}\quad\left[\because\int\frac{f'(x)}{f(x)}dx=\log|f(x)|\right]$$

Let $I=\int\frac{dx}{x^2-2x-5}=\int\frac{dx}{(x^2-2x+1)-6}=\int\frac{dx}{(x-1)^2-(\sqrt{6})^2}$ **2 Marks**

$$=\frac{1}{2\sqrt{6}}\log\left(\frac{x-1-\sqrt{6}}{x-1+\sqrt{6}}\right)\qquad\left[\because\int\frac{1}{x^2-a^2}dx=\frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|+c\right]$$

From (2), $\int\frac{x+3}{x^2-2x-5}dx=\frac{1}{2}\log\left|x^2-2x-5\right|+\frac{4}{2\sqrt{6}}\log\left|\frac{x-1-\sqrt{6}}{x-1+\sqrt{6}}\right|+C$

$$=\frac{1}{2}\log\left|x^2-2x-5\right|+\frac{2}{\sqrt{6}}\log\left|\frac{x-1-\sqrt{6}}{x-1+\sqrt{6}}\right|+C$$ **2Marks**

Mahesh

☺ FORMULAE BOX ☺

i) $\int\frac{1}{x^2+a^2}dx=\frac{1}{a}\left(\tan^{-1}\frac{x}{a}\right)+c$ ii) $\int\frac{1}{x^2-a^2}dx=\frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|+c$

iii) $\int\frac{1}{\sqrt{x^2+a^2}}dx=\log|x+\sqrt{x^2+a^2}|+c$ iv) $\int\sqrt{a^2-x^2}dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\text{Sin}^{-1}\frac{x}{a}+c$

v) $\int\frac{f'(x)}{f(x)}dx=\log|f(x)|+c$ vi) $\int\frac{f'(x)}{\sqrt{f(x)}}dx=2\sqrt{f(x)}+c$ vii) $\int f'(x)\sqrt{f(x)}dx=\frac{2}{3}\left[f(x)\right]^{3/2}$

46. Find the integral of $\frac{6x+7}{\sqrt{(x-5)(x-4)}}$

Sol: We have $\frac{6x+7}{\sqrt{(x-5)(x-4)}}=\frac{6x+7}{\sqrt{x^2-9x+20}}$

Let $6x+7=A\frac{d}{dx}(x^2-9x+20)+B \Rightarrow 6x+7=A(2x-9)+B$

Equating the coefficients of x and constant term,we get 2 Mar

$2A=6\Rightarrow A=3;\ -9A+B=7\Rightarrow B=34 \qquad \therefore 6x+7=3(2x-9)+34$

$\therefore\int\frac{6x+7}{\sqrt{x^2-9x+20}}=\int\frac{3(2x-9)+34}{\sqrt{x^2-9x+20}}dx \quad =3\int\frac{(2x-9)}{\sqrt{x^2-9x+20}}dx+34\int\frac{1}{\sqrt{x^2-9x+20}}dx$

Let $I_1=\int\frac{2x-9}{\sqrt{x^2-9x+20}}dx$ and $I_2=\int\frac{1}{\sqrt{x^2-9x+20}}dx$

$\therefore\int\frac{6x+7}{\sqrt{x^2-9x+20}}=3I_1+34I_2$.....(1)

2 Mar

First we find $I_1=\int\frac{2x-9}{\sqrt{x^2-9x+20}}dx$ Let $x^2-9x+20=t\Rightarrow(2x-9)dx=dt$

$\therefore I_1=\int\frac{dt}{\sqrt{t}}\Rightarrow I_1=2\sqrt{t}\Rightarrow I_1=2\sqrt{x^2-9x+20}$.....(2) Now $I_2=\int\frac{1}{\sqrt{x^2-9x+20}}dx$

$x^2-9x+20=x^2-9x+20+\frac{81}{4}-\frac{81}{4}$

$x^2-9x+20+\frac{81}{4}-\frac{81}{4}=\left(x-\frac{9}{2}\right)^2-\frac{1}{4}=\left(x-\frac{9}{2}\right)^2-\left(\frac{1}{2}\right)^2$

$\Rightarrow I_2=\int\frac{1}{\left(x-\frac{9}{2}\right)^2-\left(\frac{1}{2}\right)^2}dx=\log\left|\left(x-\frac{9}{2}\right)+\sqrt{x^2-9x+20}\right|$.....(3) 2 Mar

Substituting equations (2)and (3)in (1), we get

$\int\frac{6x+7}{\sqrt{x^2-9x+20}}dx=3\left[2\sqrt{x^2-9x+20}\right]+34\log\left[\left(x-\frac{9}{2}\right)+\sqrt{x^2-9x+20}\right]+C$

$=6\sqrt{x^2-9x+20}+34\log\left[\left(x-\frac{9}{2}\right)+\sqrt{x^2-9x+20}\right]+C$ 2 Mar

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47. Find the integral of $\frac{x+2}{\sqrt{x^2+2x+3}}$

Sol: $\int\frac{x+2}{\sqrt{x^2+2x+3}}dx=\frac{1}{2}\int\frac{2(x+2)}{\sqrt{x^2+2x+3}}dx=\frac{1}{2}\int\frac{2x+4}{\sqrt{x^2+2x+3}}dx$

Mahesh

$$=\frac{1}{2}\int\frac{2x+2}{\sqrt{x^2+2x+3}}dx+\frac{1}{2}\int\frac{2}{\sqrt{x^2+2x+3}}dx=\frac{1}{2}\int\frac{2x+2}{\sqrt{x^2+2x+3}}dx+\int\frac{1}{\sqrt{x^2+2x+3}}dx$$

4 Marks

$$=\frac{1}{2}\left[2\sqrt{x^2+2x+3}\right]+\int\frac{1}{\sqrt{x^2+2x+3}}dx \quad .....(2) \qquad \left[\because \int\frac{f'(x)}{\sqrt{f(x)}}dx=2\sqrt{f(x)}+c\right]$$

Let $I=\int\frac{1}{\sqrt{x^2+2x+3}}dx=\int\frac{1}{\sqrt{(x^2+2x+1)+2}}dx$

$$=\int\frac{1}{\sqrt{(x+1)^2+(\sqrt{2})^2}}dx=\log\left|(x+1)+\sqrt{x^2+2x+3}\right| \left[\because \int\frac{1}{\sqrt{x^2+a^2}}dx=\log\left(x+\sqrt{x^2+a^2}\right)+c\right]$$

From (2), $\int\frac{x+2}{\sqrt{x^2+2x+3}}dx=\sqrt{x^2+2x+3}+\log\left|(x+1)+\sqrt{x^2+2x+3}\right|+C$

4 Marks

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48. Find the integral of $\frac{5x+3}{\sqrt{x^2+4x+10}}$

Sol: Let $5x+3=A\frac{d}{dx}(x^2+4x+10)+B \Rightarrow 5x+3=A(2x+4)+B$ .....(1)

Equating the coefficients of x on both sides, we get $2A=5\Rightarrow A=5/2$

Equating the constant terms, we get $4A+B=3\Rightarrow B=3-4A=3-4\left(\frac{5}{2}\right)=3-10=-7$

2 Marks

Hence from (1), $5x+3=\frac{5}{2}(2x+4)-7$

$$\therefore\int\frac{5x+3}{\sqrt{x^2+4x+10}}dx=\int\frac{\frac{5}{2}(2x+4)-7}{\sqrt{x^2+4x+10}}dx$$

$$=\frac{5}{2}\int\frac{2x+4}{\sqrt{x^2+4x+10}}dx-7\int\frac{1}{\sqrt{x^2+4x+10}}dx$$

2 Marks

$$=\frac{5}{2}\left[2\sqrt{x^2+4x+10}\right]-7\int\frac{1}{\sqrt{x^2+4x+10}}dx \quad .....(2) \quad \left[\because \int\frac{f'(x)}{\sqrt{f(x)}}dx=2\sqrt{f(x)}+c\right]$$

$$\text{Let } I=\int\frac{1}{\sqrt{x^2+4x+10}}dx=\int\frac{1}{\sqrt{(x^2+4x+4)+6}}dx=\int\frac{1}{\sqrt{(x+2)^2+(\sqrt{6})^2}}dx$$

$$=\log\left|(x+2)+\sqrt{x^2+4x+10}\right| \qquad \left[\because \int\frac{1}{\sqrt{x^2+a^2}}dx=\log\left(x+\sqrt{x^2+a^2}\right)\right]$$

From (2),

$$\int\frac{5x+3}{\sqrt{x^2+4x+10}}dx=5\sqrt{x^2+4x+10}-7\log\left|(x+2)+\sqrt{x^2+4x+10}\right|+C$$

4 Marks

Mahesh

7 DIFERENTIAL EQUATIONS

8 Marks

PROBLEMS ON LINEAR DIFFERENTIAL EQUATIONS

49. Solve the differential equation $(\tan^{-1}y - x)\,dy = (1 + y^2)\,dx$.

Sol: Given D.E is written as $\frac{dx}{dy} = \frac{\tan^{-1}y - x}{1+y^2} \Rightarrow \frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1}y}{1+y^2}$ .....(1)

(1) is a L.D.E in x of the form $\frac{dx}{dy} + Px = Q$. Here $P = \frac{1}{1+y^2}$ and $Q = \frac{\tan^{-1}y}{1+y^2}$ 1 Mark

$\therefore IF = e^{\int P dx} = e^{\int \frac{1}{1+y^2}dy} = e^{\tan^{-1}y}$ 1 Mark

G.S is $x(IF) = \int (Q \times IF)dy + C \Rightarrow xe^{\tan^{-1}y} = \int \left(\frac{\tan^{-1}y}{1+y^2}\right) e^{\tan^{-1}y} dy + C$....(2)

Put $\tan^{-1}y = t$ then $\left(\frac{1}{1+y^2}\right)dy = dt$ 2 Marks

$\therefore I = \int te^t dt = te^t - \int 1\, e^t dt = te^t - e^t = e^t(t-1) \Rightarrow I = e^{\tan^{-1}y}(\tan^{-1}y - 1)$ .....(3)

From (2) & (3), we have $xe^{\tan^{-1}y} = e^{\tan^{-1}y}(\tan^{-1}y - 1) + C \Rightarrow x = (\tan^{-1}y - 1) + Ce^{-\tan^{-1}y}$

☞ **Except this question Q 49, the remaining Q 50 to Q 55 are L.D.E in y.** 4 Marks

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50. Find the general solution of $(1+x^2)dy + 2xydx = \cot x dx,\ (x \neq 0)$

Sol: Given DE is $(1+x^2)dy + 2xydx = \cot x dx \Rightarrow \frac{dy}{dx} + \frac{2xy}{(1+x^2)} = \frac{\cot x}{(1+x^2)}$ .....(1) 1 Mark

(1) is a L.D.E in y of the form $\frac{dy}{dx} + Py = Q$. Here $P = \frac{2x}{(1+x^2)}$ and $Q = \frac{\cot x}{(1+x^2)}$ 1 Mark

$\therefore IF = e^{\int P dx} = e^{\int \frac{2x}{1+x^2}dx} = e^{\log(1+x^2)} = 1 + x^2$

General solution is $y(IF) = \int (Q \times IF)dx + C$ 2 Marks

$\therefore y(1+x^2) = \int \left(\frac{\cot x}{1+x^2} \times (1+x^2)\right)dx + C = \int \cot x dx + C$

$\Rightarrow y(1+x^2) = \log|\sin x| + C$ 4 Marks

Mahesh

51. Find the general solution of $x\frac{dy}{dx}+y-x+xy\cot x=0,\ (x\neq 0)$

Sol: Given D.E is $x\frac{dy}{dx}+y-x+xy\cot x=0 \Rightarrow \frac{dy}{dx}+\frac{y}{x}-1+y\cot x=0$

$\Rightarrow \frac{dy}{dx}+y\left(\frac{1}{x}+\cot x\right)-1=0 \Rightarrow \frac{dy}{dx}+\left(\frac{1}{x}+\cot x\right)y=1$ .....(1) **1 Mark**

(1) is a L.D.E in y of the form $\frac{dy}{dx}+Py=Q$. Here $P=\left(\frac{1}{x}+\cot x\right)$ and $Q=1$ **1 Mark**

$IF=e^{\int P dx}=e^{\int\left(\frac{1}{x}+\cot x\right)dx}=e^{\log x+\log(\sin x)}=e^{\log(x\sin x)}=x\sin x$

General solution is $y(IF)=\int(Q\times IF)dx+C=\int(1\times x\sin x)dx+C$ **2 Marks**

$=\int(x\sin x)dx+C=x\int\sin x dx-\int\left[\frac{d}{dx}(x).\int\sin x dx\right]+C$

$=x(-\cos x)-\int 1.(-\cos x)dx+C=-x\cos x+\sin x+C$

$\Rightarrow y=\frac{-x\cos x}{x\sin x}+\frac{\sin x}{x\sin x}+\frac{C}{x\sin x} \Rightarrow y=-\cot x+\frac{1}{x}+\frac{C}{x\sin x} \Rightarrow y=\frac{1}{x}-\cot x+\frac{C}{x\sin x}$

4 Marks

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52. Find the particular solution of the differential equation

$\frac{dy}{dx}+2y\tan x=\sin x$; $y=0$ when $x=\frac{\pi}{3}$

Sol: Given D.E is $\frac{dy}{dx}+2y\tan x=\sin x$ .....(1) **1 Mark**

(1) is a L.D.E in y of the form $\frac{dy}{dx}+Py=Q$. Here $P=2\tan x$ and $Q=\sin x$ **1 Mark**

$IF=e^{\int P dx}=e^{\int 2\tan x dx}=e^{2\log|\sec x|}=e^{\log(\sec^2 x)}=\sec^2 x$

General solution is $y(IF)=\int(Q\times IF)dx+C \Rightarrow y(\sec^2 x)=\int(\sin x.\sec^2 x)dx+C$ **2 Marks**

$\therefore y\sec^2 x=\int(\sec x.\tan x)dx+C \Rightarrow y\sec^2 x=\sec x+C$ .....(2)

Now $y=0$ at $x=\frac{\pi}{3} \Rightarrow 0\times\sec^2\frac{\pi}{3}=\sec\frac{\pi}{3}+C \Rightarrow 0=2+C \Rightarrow C=-2$

From (2) $y\sec^2 x=\sec x-2 \Rightarrow y=\cos x-2\cos^2 x$, which is the required particular solution.

4 Marks

Mahesh

53. **Find the particular solution of the differential equation**

$\frac{dy}{dx} + y\cot x = 2x + x^2\cot x (x \neq 0)$ **given that** $y = 0$ **when** $x = \frac{\pi}{2}$

Sol: Given D.E is $\frac{dy}{dx} + y\cot x = 2x + x^2\cot x$ .....(1)

(1) is a L.D.E in y of the form $\frac{dy}{dx} + Py = Q$. Here $P = \cot x$ and $Q = 2x + x^2\cot x$.

$\therefore IF = e^{\int Pdx} = e^{\int \cot x dx} = e^{\log \sin x} = \sin x$

General solution is $y(IF) = \int (Q \times IF)dx + C$

$y.\sin x = \int (2x + x^2\cot x)\sin x dx + C \Rightarrow y\sin x = \int 2x\sin x dx + \int x^2\cos x dx + C$

$\Rightarrow y\sin x = \sin x\left(\frac{2x^2}{2}\right) - \int \cos x\left(\frac{2x^2}{2}\right)dx + \int x^2\cos x dx + C$

$\Rightarrow y\sin x = x^2\sin x - \int x^2\cos x dx + \int x^2\cos x dx + C$

$\Rightarrow y\sin x = x^2\sin x + C$.....(2)

Now $y = 0$ and $x = \frac{\pi}{2}$ in (2), we get $0 = \left(\frac{\pi}{2}\right)^2 \sin\left(\frac{\pi}{2}\right) + C \Rightarrow C = \frac{-\pi^2}{4}$

From (2), $y\sin x = x^2\sin x - \frac{\pi^2}{4} \Rightarrow y = x^2 - \frac{\pi^2}{4\sin x}$

This is the required particular solution of the given D.E

54. **Find the particular solution of the differential equation**

$\frac{dy}{dx} - 3y\cot x = \sin 2x$; $y = 2$ **when** $x = \frac{\pi}{2}$

Sol: Given D.E is $\frac{dy}{dx} - 3y\cot x = \sin 2x$ .....(1)

(1) is a L.D.E in y of the form $\frac{dy}{dx} + Py = Q$ Here $P = -3\cot x$ and $Q = \sin 2x$

$IF = e^{\int Pdx} = e^{-3\int \cot x dx} = e^{-3\log|\sin x|} = e^{\log|\sin x|^{-3}} = e^{\log\left|\frac{1}{\sin^3 x}\right|} = \frac{1}{\sin^3 x}$

General solution is $y(IF) = \int (Q \times IF)dx + C$

$\therefore y.\frac{1}{\sin^3 x} = \int\left[\sin 2x.\frac{1}{\sin^3 x}\right]dx + C$

Mahesh

$\Rightarrow y\cos ec^3x = 2\int(\cot x \cos ecx)dx + C \quad \Rightarrow y\cos ec^3x = -2\cos ecx + C$

$\Rightarrow y = \frac{-2}{\cos ec^2x} + \frac{C}{\cos ec^3x} \Rightarrow y = -2\sin^2 x + C\sin^3 x$ ....(2)

Now $y = 2$ at $x = \frac{\pi}{2} \Rightarrow 2 = -2 + C \Rightarrow C = 4$

$\therefore$ From (2), $y = -2\sin^2 x + 4\sin^3 x \Rightarrow y = 4\sin^3 x - 2\sin^2 x$

This is the required particular solution of the given D.E. **4 Marks**

Find the equation of a curve passing through the origin given that the slope of the tangent to the curve at any point (x, y) is equal to the sum of the coordinates of the point. [BMP]

Let F(x, y)be the curve passing through origin.

At (x, y), slope of curve will be $\frac{dy}{dx}$

$\therefore \frac{dy}{dx} = x + y \Rightarrow \frac{dy}{dx} - y = x$ .....(1) **1 Mark**

(1) is a L.D.E in y of the form $\frac{dy}{dx} + Py = Q$. Here $P = -1$ and $Q = x$ **1 Mark**

$\therefore IF = e^{\int Pdx} = e^{\int(-1)dx} = e^{-x}$ **1 Mark**

General solution is $y(IF) = \int(Q \times IF)dx + C$ **1 Mark**

$\therefore ye^{-x} = \int xe^{-x}dx + C \quad \Rightarrow ye^{-x} = x\int e^{-x}dx - \int\left[\frac{d}{dx}(x).\int e^{-x}dx\right]dx + C$

$\Rightarrow ye^{-x} = -xe^{-x} + \int e^{-x}dx + C$

$\Rightarrow ye^{-x} = -xe^{-x} + (-e^{-x}) + C \Rightarrow ye^{-x} = -e^{-x}(x+1) + C$

$\Rightarrow x + y + 1 = Ce^{-x}$.....(2)

As the curve passes through origin O(0,0), from (2) we have 0+0+1=C.e $\Rightarrow C = 1$

(2) $\Rightarrow x + y + 1 = e^x$ which is the required equation of the curve. **4 Marks**

Mahesh

PROBLEMS ON HOMOGENEOUS D.E

56. Show that the differential equation $(x^2 - y^2)\,dx + 2xy\,dy = 0$ is homogeneous and solve it.

Sol: Given D.E is $(x^2-y^2)dx+2xydy=0 \Rightarrow 2xydy=(y^2-x^2)dx \Rightarrow \frac{dy}{dx}=\frac{(y^2-x^2)}{2xy}$ .....(1)

Let $F(x,y)=\frac{(y^2-x^2)}{2xy} \Rightarrow F(\lambda x,\lambda y)=\left[\frac{(\lambda y)^2-(\lambda x)^2}{2(\lambda x)(\lambda y)}\right]=\frac{\lambda^2(y^2-x^2)}{\lambda^2(2xy)}=\lambda^0 F(x,y)$ 2 Marks

$\therefore$ Given D.E is a homogeneous equation.

Let $y=vx \Rightarrow \frac{dy}{dx}=v+x\frac{dv}{dx}$ 1 Mark

From (1), $v+x\frac{dv}{dx}=\left[\frac{(vx)^2-x^2}{2x(vx)}\right]=\frac{v^2-1}{2v}$ 1 Mark

$\Rightarrow x\frac{dv}{dx}=\frac{v^2-1}{2v}-v=\frac{v^2-1-2v^2}{2v}=\frac{-(1+v^2)}{2v}$ $\quad \therefore x\frac{dv}{dx}=\frac{-(1+v^2)}{2v}$

$\Rightarrow \frac{2v}{1+v^2}dv=-\frac{dx}{x} \Rightarrow \log(1+v^2)=-\log x+\log C=\log\frac{C}{x}$

$\Rightarrow 1+v^2=\frac{C}{x} \Rightarrow \left[1+\frac{y^2}{x^2}\right]=\frac{C}{x} \Rightarrow x^2+y^2=Cx$

This is the required general soltuion to the given D.E. 4 Marks

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57. Show that the differential equation $(x^2 + xy)\,dy = (x^2 + y^2)\,dx$ is homogeneous and solve it.

Sol: Given D.E is $(x^2+xy)dy=(x^2+y^2)dx \Rightarrow \frac{dy}{dx}=\frac{x^2+y^2}{x^2+xy}$ .....(1)

Let $F(x,y)=\frac{x^2+y^2}{x^2+xy} \Rightarrow F(\lambda x,\lambda y)=\frac{(\lambda x)^2+(\lambda y)^2}{(\lambda x)^2+(\lambda x)(\lambda y)}=\frac{\lambda^2(x^2+y^2)}{\lambda^2(x^2+xy)}=\lambda^0 F(x,y)$

$\therefore$ Given D.E is a homogeneous equation.

Let $y=vx \Rightarrow \frac{dy}{dx}=v+x\frac{dv}{dx}$

2 Marks

Mahesh

From (1), $v+x\frac{dv}{dx}=\frac{x^2+(vx)^2}{x^2+x(vx)}=\frac{1+v^2}{1+v}$

$\Rightarrow x\frac{dv}{dx}=\frac{1+v^2}{1+v}-v=\frac{(1+v^2)-v(1+v)}{1+v}=\frac{1-v}{1+v}$ **2 Marks**

$\Rightarrow\left(\frac{1+v}{1-v}\right)dv=\frac{dx}{x}\Rightarrow\left(\frac{2-1+v}{1-v}\right)dv=\frac{dx}{x}\Rightarrow\left(\frac{2}{1-v}-1\right)dv=\frac{dx}{x}$

$\Rightarrow -2\log(1-v)-v=\log x-\log C\Rightarrow v=-2\log(1-v)-\log x+\log C$

$\Rightarrow v=\log\left[\frac{c}{x(1-v)^2}\right]\Rightarrow\frac{y}{x}=\log\left[\frac{c}{x\left(1-\frac{y}{x}\right)^2}\right]\Rightarrow\frac{y}{x}=\log\left[\frac{Cx}{(x-y)^2}\right]$

$\Rightarrow\frac{Cx}{(x-y)^2}=e^{\frac{y}{x}}\Rightarrow(x-y)^2=Cxe^{\frac{-y}{x}}$

This is the required general soltuion to the given D.E. **4 Marks**

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58. Show that the differential equation $xdy-ydx=\sqrt{x^2+y^2}dx$ is homogeneous and solve it.

Sol: Given D.E is $xdy-ydx=\sqrt{x^2+y^2}dx\Rightarrow xdy=ydx+\sqrt{x^2+y^2}dx$

$\Rightarrow xdy=(y+\sqrt{x^2+y^2})dx\Rightarrow\frac{dy}{dx}=\frac{y+\sqrt{x^2+y^2}}{x}=F(x,y)$ ...(1).

Let $F(x,y)=\frac{y+\sqrt{x^2+y^2}}{x}\Rightarrow F(\lambda x,\lambda y)=\frac{\lambda y+\sqrt{(\lambda x)^2+(\lambda y)^2}}{\lambda x}=\lambda\left(\frac{y+\sqrt{x^2+y^2}}{\lambda x}\right)=\lambda^0F(x,y)$ **2 Marks**

$\therefore$ Given D.E is a homogeneous equation.

Let $y=vx\Rightarrow\frac{dy}{dx}=v+x\frac{dv}{dx}$ **1 Mark**

From (1), $v+x\frac{dv}{dx}=\frac{vx+\sqrt{x^2+v^2x^2}}{x}=\frac{x(v+\sqrt{1+v^2})}{x}=v+\sqrt{1+v^2}$ **1 Mark**

$\Rightarrow x\frac{dv}{dx}=v+\sqrt{1+v^2}-v=\sqrt{1+v^2}\Rightarrow\frac{dv}{\sqrt{1+v^2}}=\frac{dx}{x}\Rightarrow\int\frac{dv}{\sqrt{1+v^2}}=\int\frac{dx}{x}$

$\Rightarrow\log(v+\sqrt{1+v^2})=\log x+\log c\Rightarrow v+\sqrt{1+v^2}=cx\Rightarrow\frac{y}{x}+\sqrt{1+\frac{y^2}{x^2}}=cx\Rightarrow y+\sqrt{x^2+y^2}=cx^2$

This is the required general soltuion to the given D.E. **4 Marks**

Mahesh

59. Find the particular solution of $x^2\,dy + (xy + y^2)\,dx = 0$; $y = 1$ when $x = 1$

Sol: Given D.E is $x^2dy + (xy + y^2)dx = 0 \Rightarrow x^2dy = -(xy + y^2)dx \Rightarrow \frac{dy}{dx} = \frac{-(xy + y^2)}{x^2}$ ...(1)

It is clear that (1) is a homogeneous D.E

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

From (1), $v + x\frac{dv}{dx} = \frac{-[x.vx + (vx)^2]}{x^2} = -v - v^2$

$\Rightarrow x\frac{dv}{dx} = -v^2 - 2v = -v(v+2) \Rightarrow \frac{dv}{v(v+2)} = -\frac{dx}{x}$

$\Rightarrow \frac{1}{2}\left[\frac{2}{v(v+2)}\right]dv = -\frac{dx}{x} \Rightarrow \frac{1}{2}\left[\frac{(v+2)-v}{v(v+2)}\right]dv = -\frac{dx}{x}$

$\Rightarrow \frac{1}{2}\left[\frac{1}{v} - \frac{1}{v+2}\right]dv = -\frac{dx}{x} \Rightarrow \frac{1}{2}[\log v - \log(v+2)] = -\log x + \log C$

$\Rightarrow \frac{1}{2}\log\left(\frac{v}{v+2}\right) = \log\frac{C}{x} \Rightarrow \frac{v}{v+2} = \left(\frac{C}{x}\right)^2$

$\Rightarrow \frac{v}{v+2} = \left(\frac{C}{x}\right)^2 \Rightarrow \frac{\frac{y}{x}}{\frac{y}{x}+2} = \left(\frac{C}{x}\right)^2 \Rightarrow \frac{y}{y+2x} = \frac{C^2}{x^2} \Rightarrow \frac{x^2y}{y+2x} = C^2$ 4 ½

Also from the given data of the problem , we have

$\Rightarrow \frac{1}{1+2} = C^2 \Rightarrow C^2 = \frac{1}{3} \Rightarrow \frac{x^2y}{y+2x} = \frac{1}{3} \Rightarrow y + 2x = 3x^2y$ 1

PQ **Show that the differential equation $(x - y)\,dy - (x + y)\,dx = 0$ is homogeneous and solve it.**

Manesh

60. Show that the family of curves for which the slope of the tangent at any point (x, y) on it is $\frac{x^2+y^2}{2xy}$ given by $x^2-y^2=Cx$.

Sol: We know that the slope of the tangent at any point P(x,y) on a curve is $\frac{dy}{dx}$

$$\therefore \frac{dy}{dx}=\frac{x^2+y^2}{2xy} \quad \ldots\ldots(1)$$

It is clear that (1) is a homogeneous D.E **1 Mark**

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ **1 Mark**

From (1), $v + x\frac{dv}{dx} = \frac{1+v^2}{2v} \Rightarrow x\left(\frac{dv}{dx}\right) = \frac{1+v^2}{2v} - v = \frac{1+v^2-2v^2}{2v} \Rightarrow x\frac{dv}{dx} = \frac{1-v^2}{2v}$ **2 Marks**

$$\Rightarrow \frac{2v}{1-v^2}dv = \frac{dx}{x} \Rightarrow \frac{2v}{v^2-1}dv = -\frac{dx}{x}$$

$$\therefore \int \frac{2v}{v^2-1}dv = -\int \frac{1}{x}dx \Rightarrow \log|v^2-1| = -\log|x| + \log|C_1|$$

$$\Rightarrow \log|(v^2-1)(x)| = \log|C_1| \Rightarrow (v^2-1)x = \pm C_1$$

Replacing v by $\frac{y}{x}$, we get $\left(\frac{y^2}{x^2}-1\right)x = \pm C_1$

$\Rightarrow (y^2-x^2) = \pm C_1 x \Rightarrow x^2-y^2 = Cx$, Hence proved. **4 Marks**

Mahesh

SHINING STARS'

LAQ STAR QUESTIONS PLUS

PQ QR Code

SQP QR Code

MATRICES

1. For the matrix $A=\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & -3\\ 2 & -1 & 3\end{bmatrix}$, Show that $A^3-6A^2+5A+11I=O$. Hence, find A^{-1}.

2. If $A=\begin{bmatrix}2 & -1 & 1\\ -1 & 2 & -1\\ 1 & -1 & 2\end{bmatrix}$, Verify that $A^3-6A^2+9A-4I=O$ and hence find A^{-1}

3D-GEOMETRY

3. Find the vector equation of the line passing through the point $(1,2,-4)$ and perpendicular to the two lines: $\frac{x-8}{3}=\frac{y+19}{-16}=\frac{z-10}{7}$ and $\frac{x-15}{3}=\frac{y-29}{-8}=\frac{z-5}{-5}$

PROBABILITY

4. A and B throw a die alternatively till one of them gets a '6' and wins the game. Find their respective probabilities of winning, if A starts first.

APPLICATION OF DERIVATIVES

5. The volume of a cube is increasing at the rate of 8 cm^3/s. How fast is the surface area increasing when the length of an edge is 12 cm?

6. A balloon, which always remains spherical on inflation, is being inflated by pumping in 900 cubic centimetres of gas per second. Find the rate at which the radius of the balloon increases when the radius is 15 cm.

7. A man of height 2 metres walks at a uniform speed of 5 km/h away from a lamp post which is 6 metres high. Find the rate at which the length of his shadow increases.

8. A window is in the form of a rectangle surmounted by a semicircular opening. The total perimeter of the window is 10 m. Find the dimensions of the window to admit maximum light through the whole opening.

INTEGRATION

9. Evaluate $\int_0^{\pi}\frac{x\,dx}{a^2\cos^2x+b^2\sin^2x}$

10. Find the integral of $\int_0^{\frac{\pi}{2}}\frac{\cos^2 x\,dx}{\cos^2 x+4\sin^2 x}$

11. Evaluate $\int\frac{1}{1+\sin x+\cos x}dx$.

12. Find $\int\left[\sqrt{\cot x}+\sqrt{\tan x}\right]dx$

13. Find the integral of $\frac{(x^2+1)(x^2+2)}{(x^2+3)(x^2+4)}$

14. Find the integral of $\frac{1}{x^4-1}$

☞ Some times, some sections contain one or two unexpected, odd, difficult & different questions. Such questions are given here. You can get their answers by scanning the above QR code.

Naresh